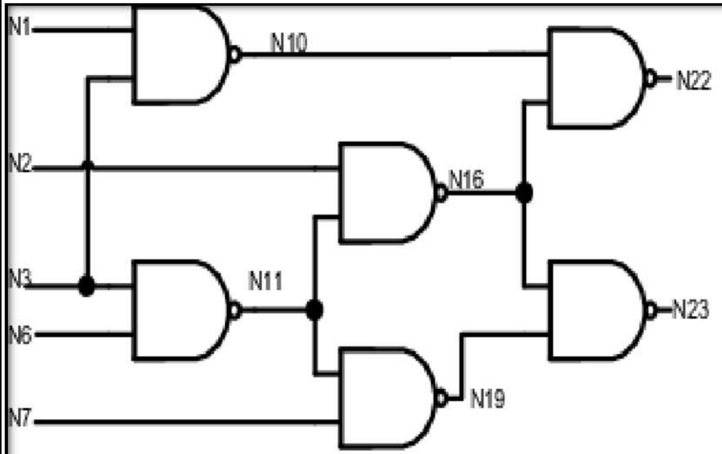
Digital Systems Design Lab: BECE102P

Model Sim Demo




```
module c17 (N1,N2,N3,N6,N7,N22,N23);
input N1,N2,N3,N6,N7;
output N22,N23;
wire N10,N11,N16,N19;
nand NAND2_1 (N10, N1, N3);
nand NAND2_2 (N11, N3, N6);
nand NAND2_3 (N16, N2, N11);
nand NAND2_4 (N19, N11, N7);
nand NAND2_5 (N22, N10, N16);
nand NAND2_6 (N23, N16, N19);
endmodule
```

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Assistant Professor

Department of Communication Engineering

School of Electronics Engineering (SENSE)

Vellore Institute of Technology, Vellore

Lab Task Discussion

Lab Task Overview

Lab Task Discussion

Lab Experiment : Model Sim

https://tinyurl.com/mvucv8yh

Lab Task Discussion

Lab Experiment : Model Sim

Language : Verilog HDL

Lab Task Discussion

Lab Experiment : Model Sim

Language : Verilog

Total Task : 5

Digital Logic Representation

- **Symbol**
- **Boolean Expression**
- **Truth table**

Introduction to Verilog

Digital Logic Representation

Symbol

Gates:
- Logic gates
- Logic Circuit

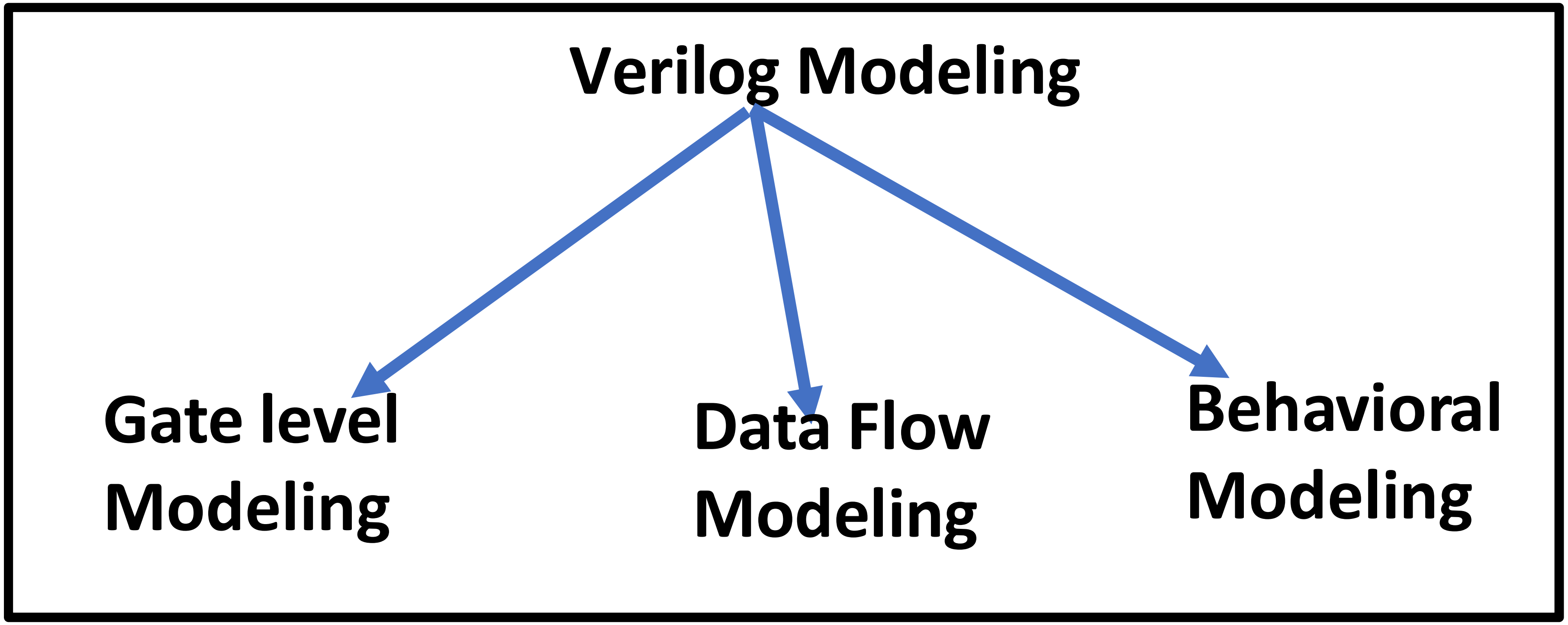
Boolean Expression

Data Flow
- Using Operator
- And variable

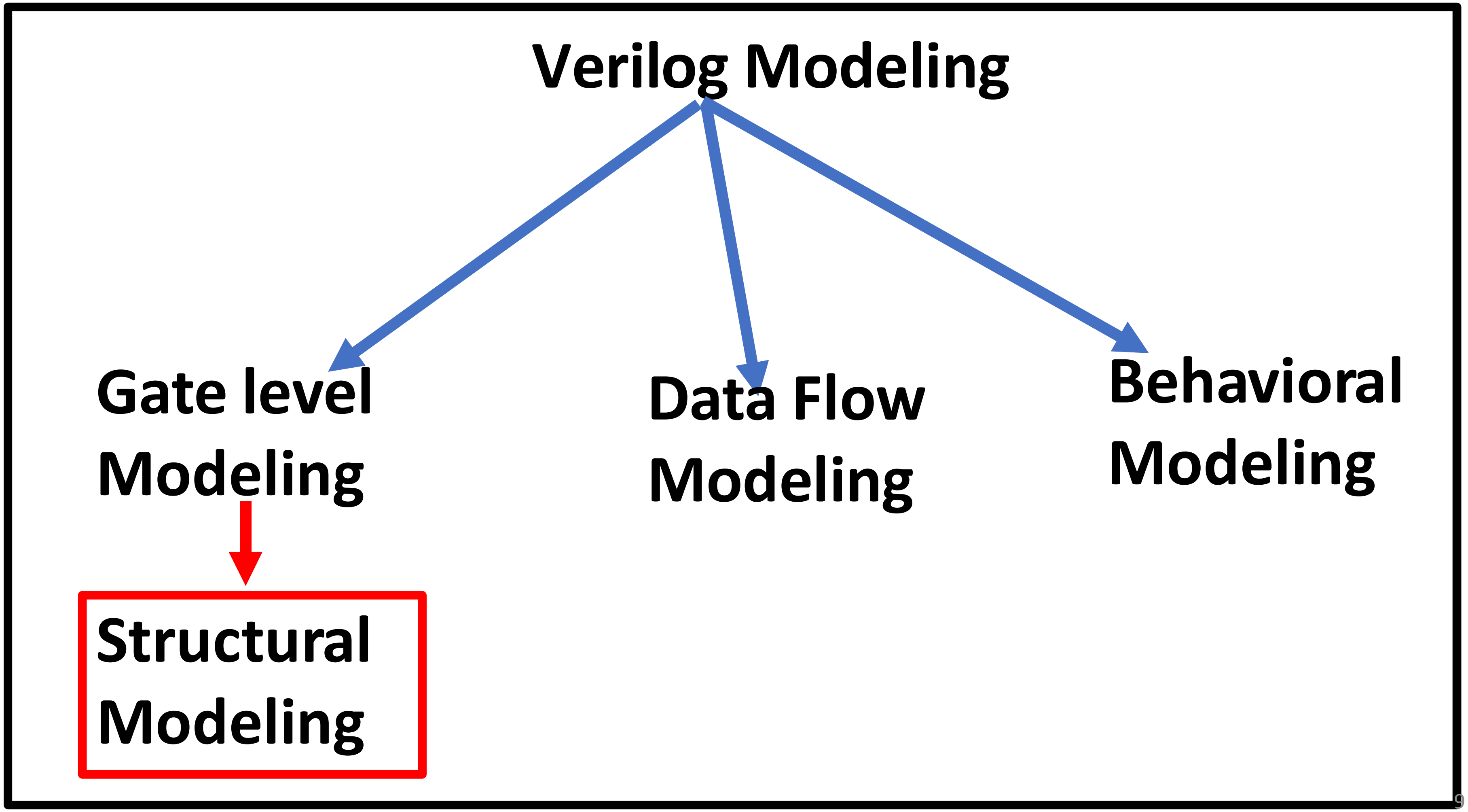
Truth table

Circuit Behavior
In terms of 1 and 0

Introduction to Verilog



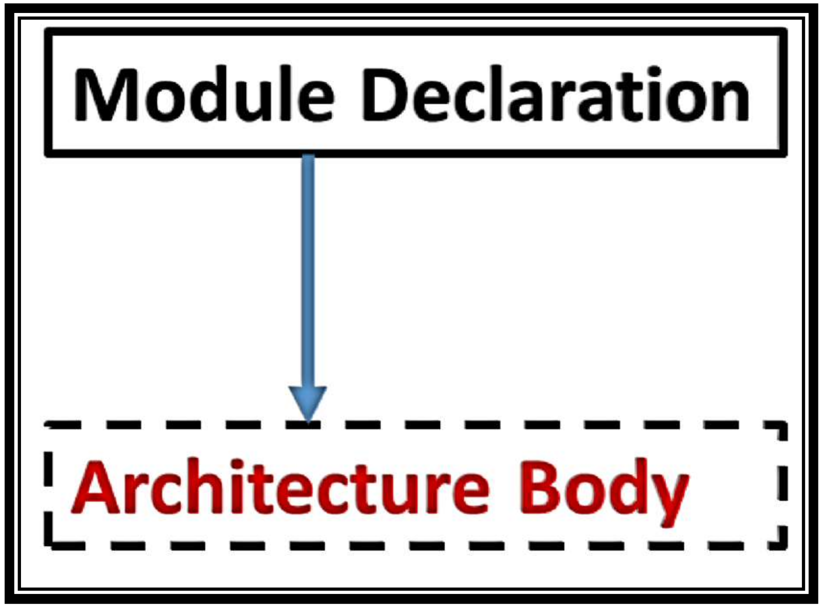
Introduction to Verilog



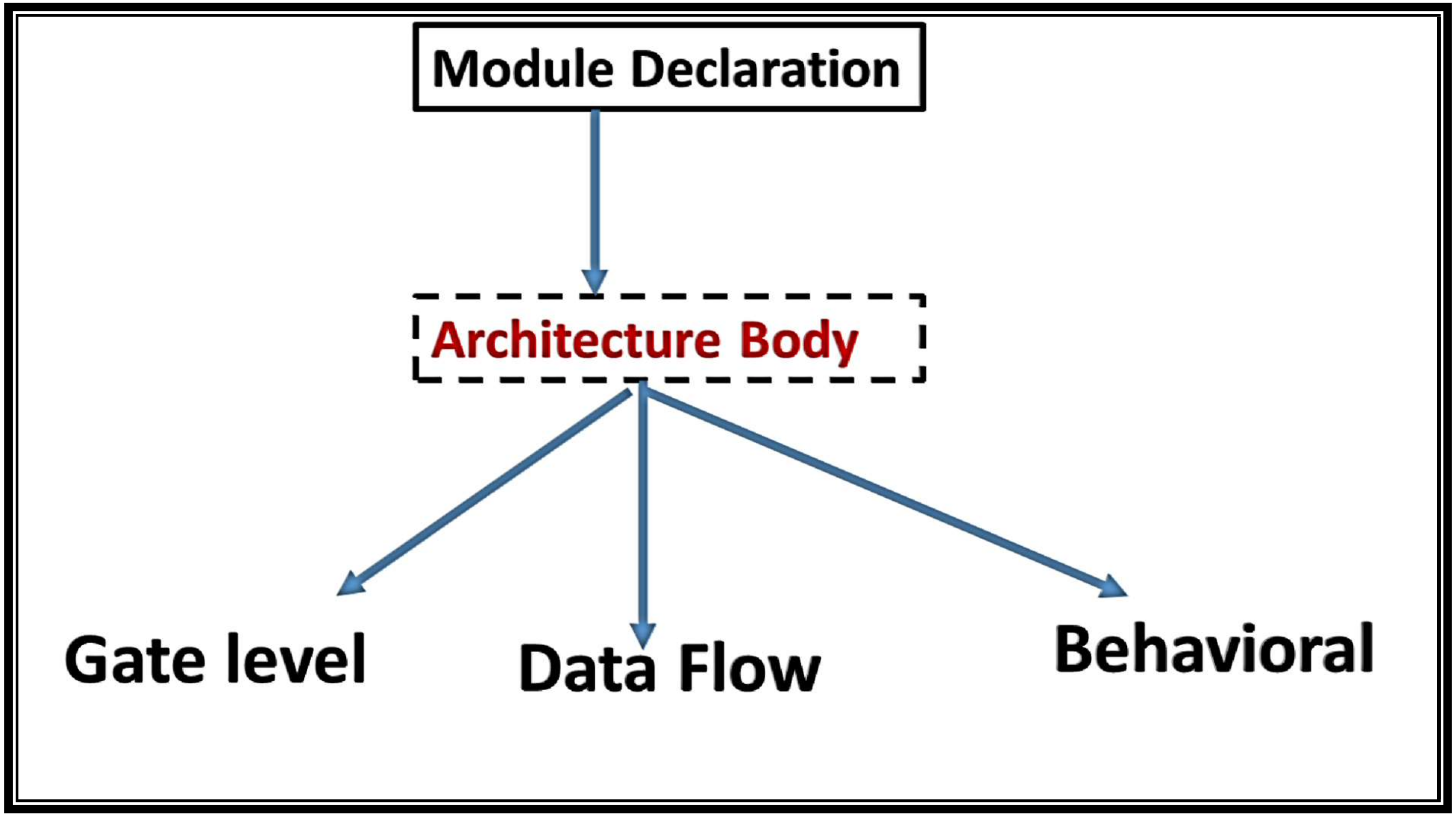
Module

What do You mean By Module?

Module



Module

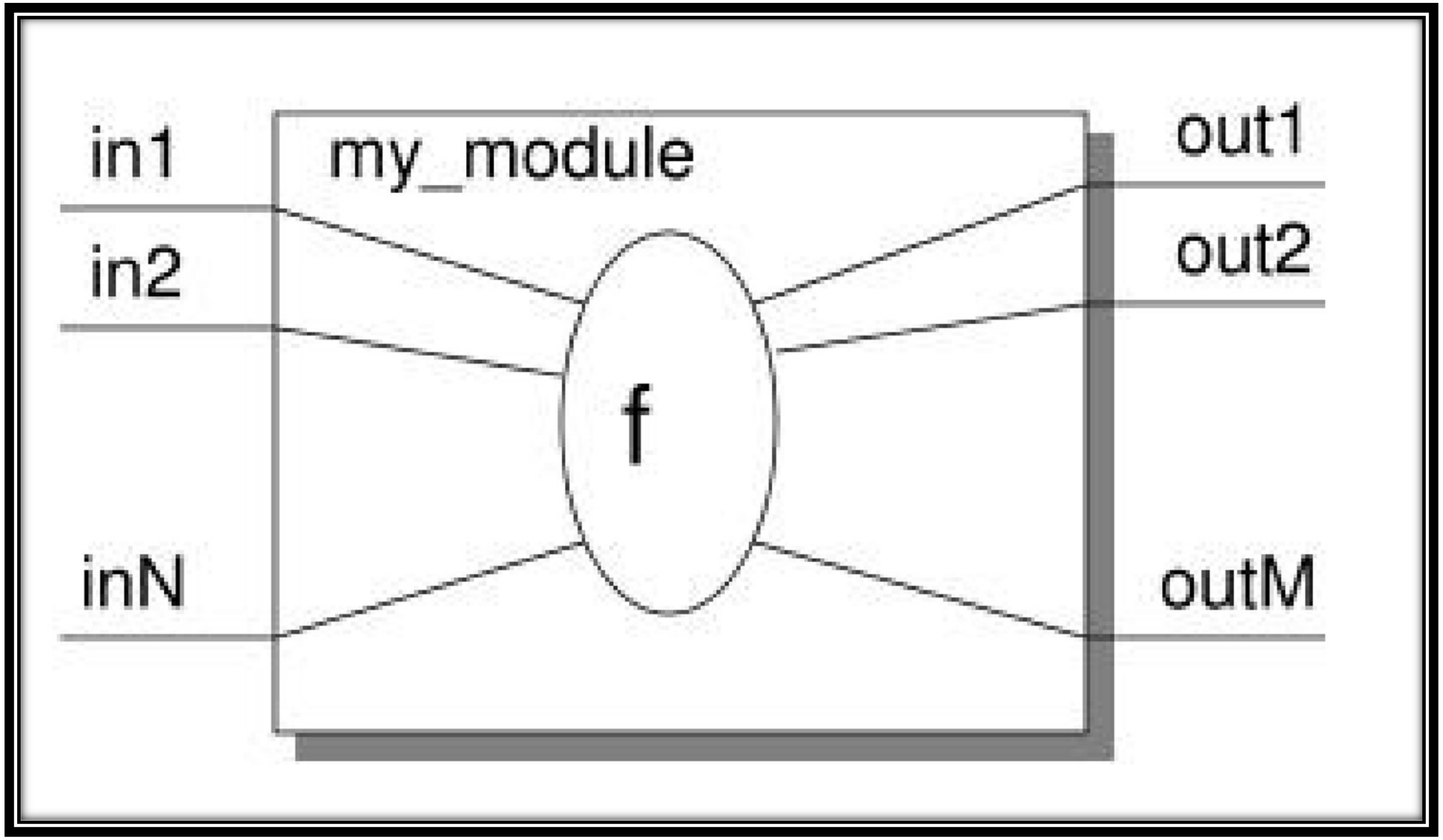


Module Declaration
Architecture Body
Gate level
Data Flow
Behavioral

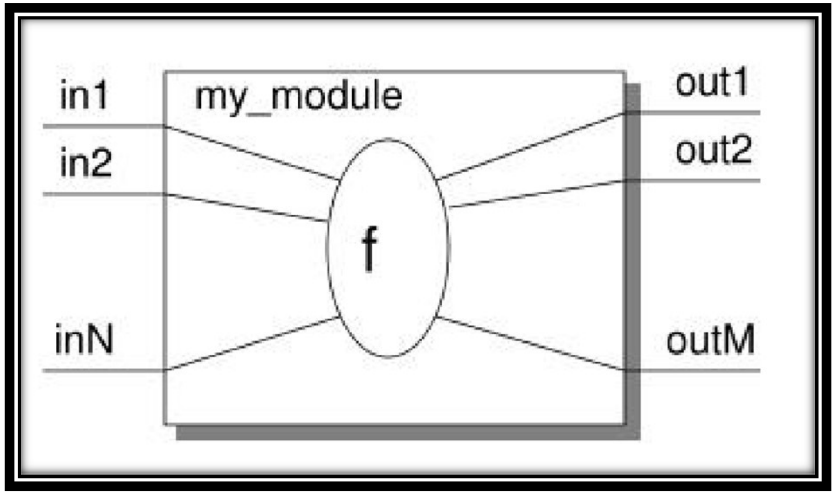
Keywords

- Note : All keywords are defined in lower case
- Examples :
- module, endmodule
- input, output, inout
- reg, integer, real, time
- not, and, nand, or, nor, xor
- parameter
- begin, end
- fork, join
- specify, endspecify

Module



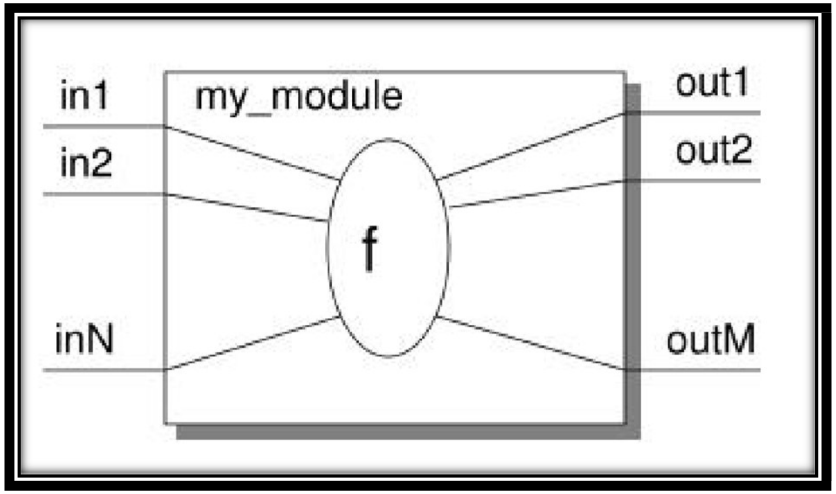
Module




```
module my_module(out1, .., inN);
output out1, .., outM;
input in1, .., inN;
```

Module Declaration

Module



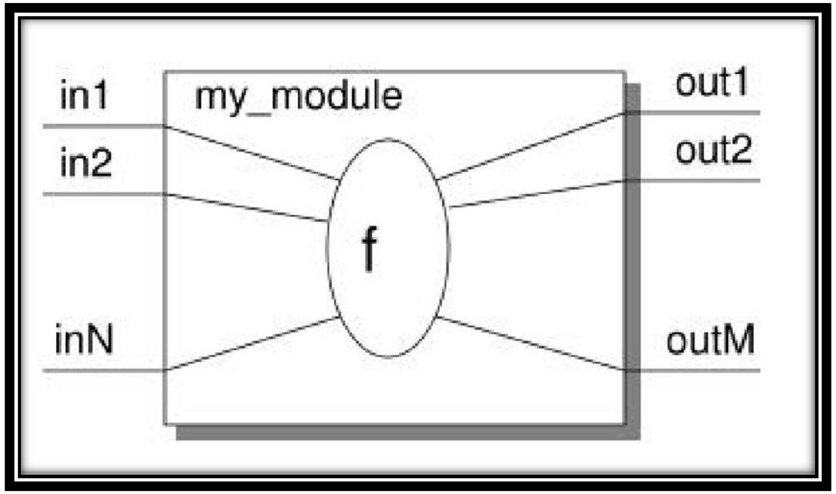

```
module my_module(out1, .., inN);
output out1, .., outM;
input in1, .., inN;

.. // declarations
.. // description of f (maybe
.. // sequential)

endmodule
```

Architecture Body

Module




```
module my_module(out1, .., inN);
output out1, .., outM;
input in1, .., inN;

.. // declarations
.. // description of f (maybe
.. // sequential)

endmodule
```

Everything you write in Verilog must be inside a module
exception: compiler directives

Gate Level Modeling

Gate Level Modeling

Gate Level Modeling

Primitives

Predefined user structures

Gate Level Modeling

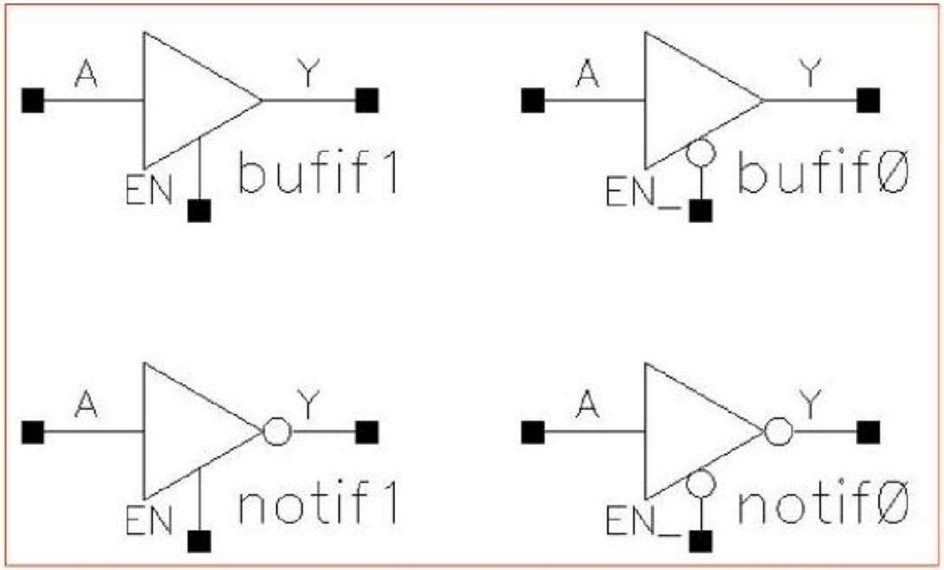
Gate Level Modeling

Primitives

Predefined user structures

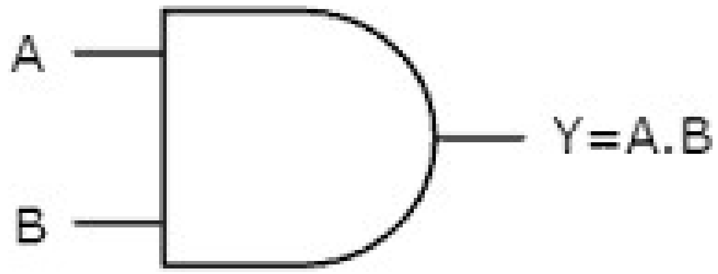
Verilog Primitives

- Basic logic gates only
  - and
  - or
  - not
  - buf
  - xor
  - nand
  - nor
  - xnor
  - bufif1, bufif0
  - notif1, notif0



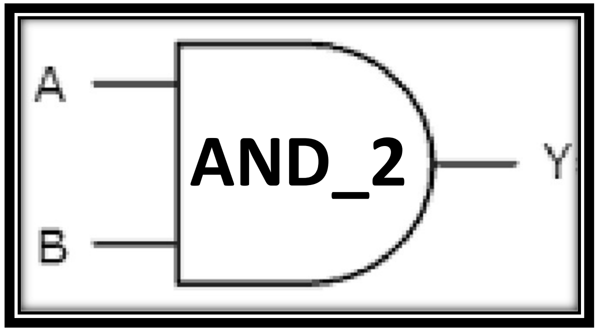
Gate Level Modeling

Logic Circuit of the AND gate

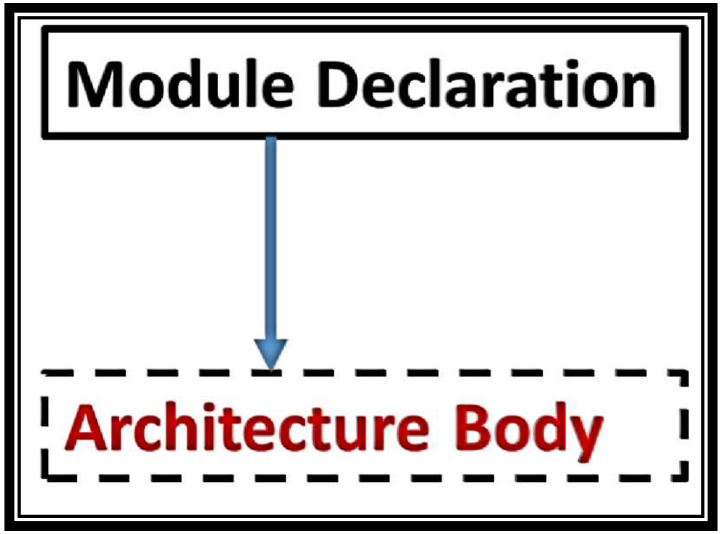


Gate Level Modeling

Logic Circuit of the AND gate

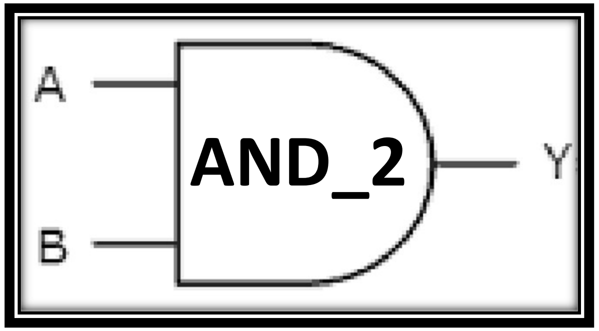


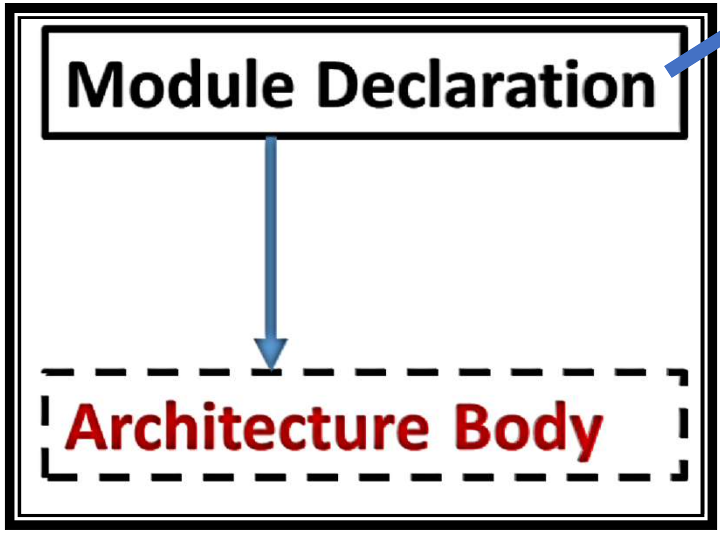
Gate-level modeling



Gate Level Modeling

Logic Circuit of the AND gate



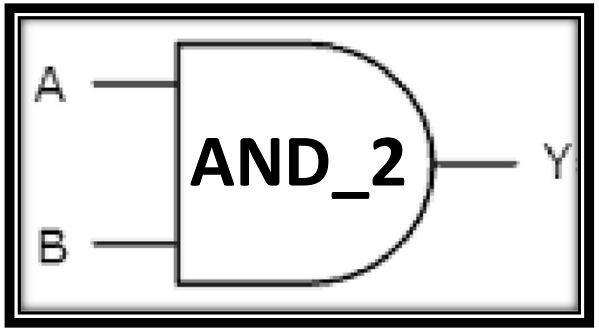


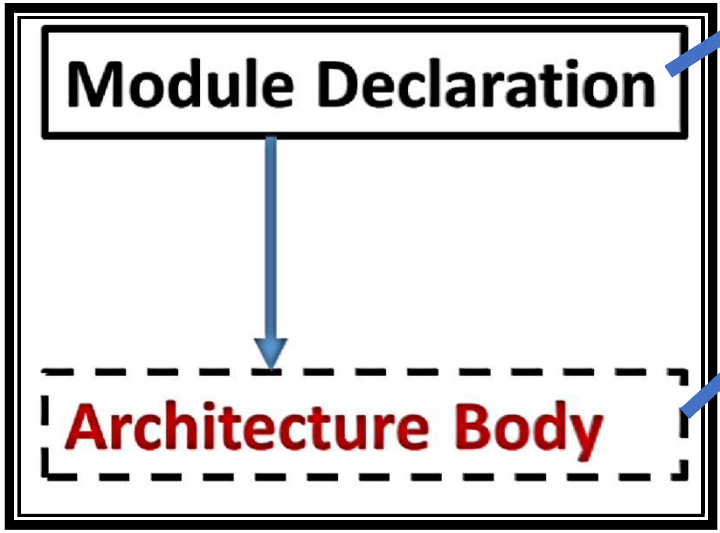
Gate-level modeling

```
module AND_2( Y, A, B);
 Input A,B;
 output Y;
```

Gate Level Modeling

Logic Circuit of the AND gate





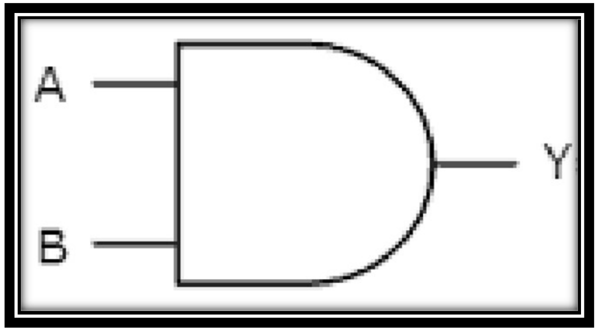
Gate-level modeling

```
module AND_2( Y, A, B);
 Input A,B;
 output Y;

and(Y, A, B);
endmodule
```

Gate Level Modeling

Logic Circuit of the AND gate



Gate-level modeling

```
module AND_2(output Y, input A, B);
   and(Y, A, B);
endmodule
```

OR

```
module AND_2( Y, A, B);
 Input A,B;
 output Y;
    and(Y, A, B);
endmodule
```

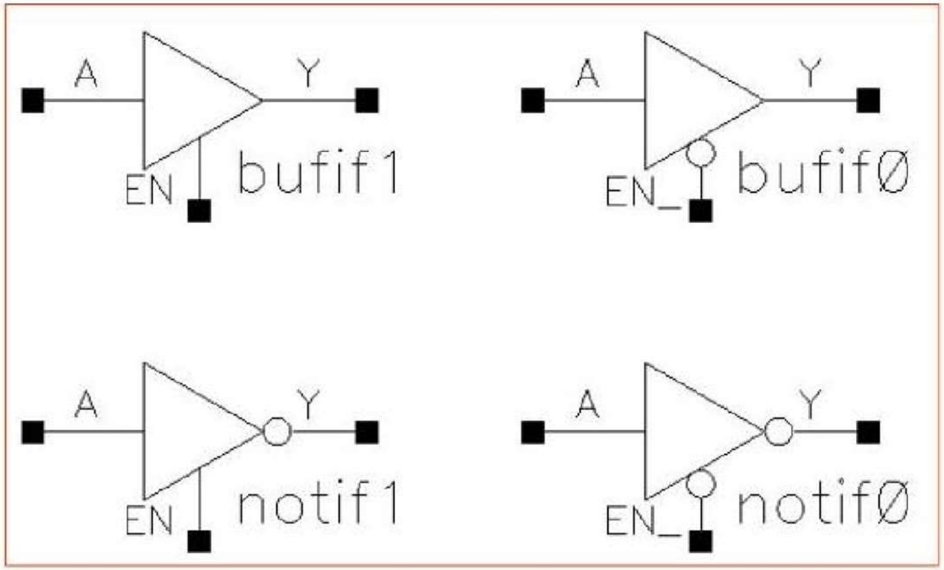
Model Sim Demo :

Example Using AND Gate using Gate Level Modeling

Gate Level Modeling

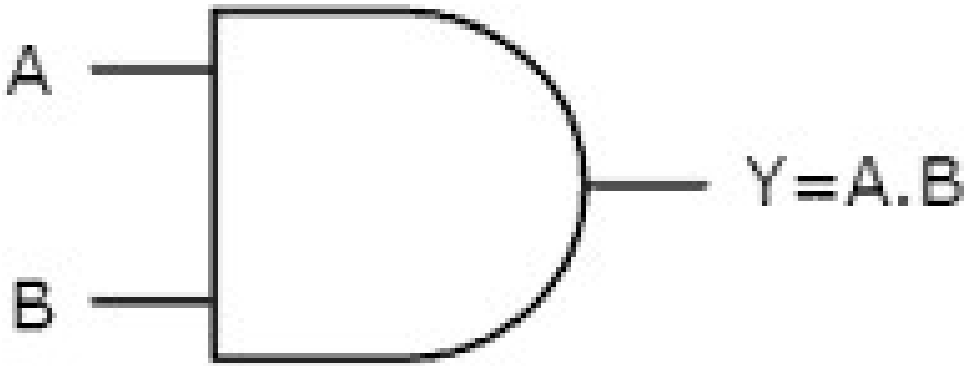
Verilog Primitives

- Basic logic gates only
  - and
  - or
  - not
  - buf
  - xor
  - nand
  - nor
  - xnor
  - bufif1, bufif0
  - notif1, notif0



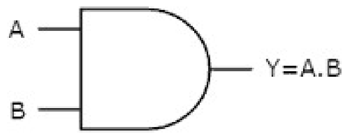
Gate Level Modeling

Logic Circuit of the AND gate



Gate Level Modeling

Logic Circuit of the AND gate

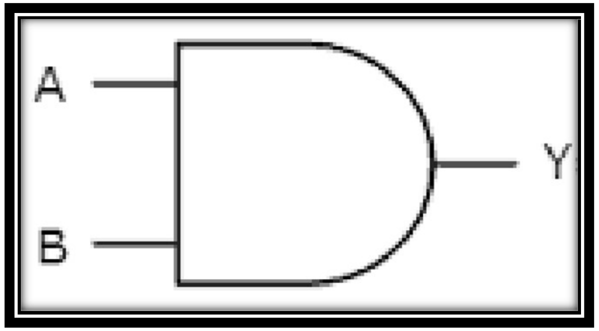


Gate-level modeling

```
module AND_2(output Y, input A, B);
   and(Y, A, B);
endmodule
```

Gate Level Modeling

Logic Circuit of the AND gate

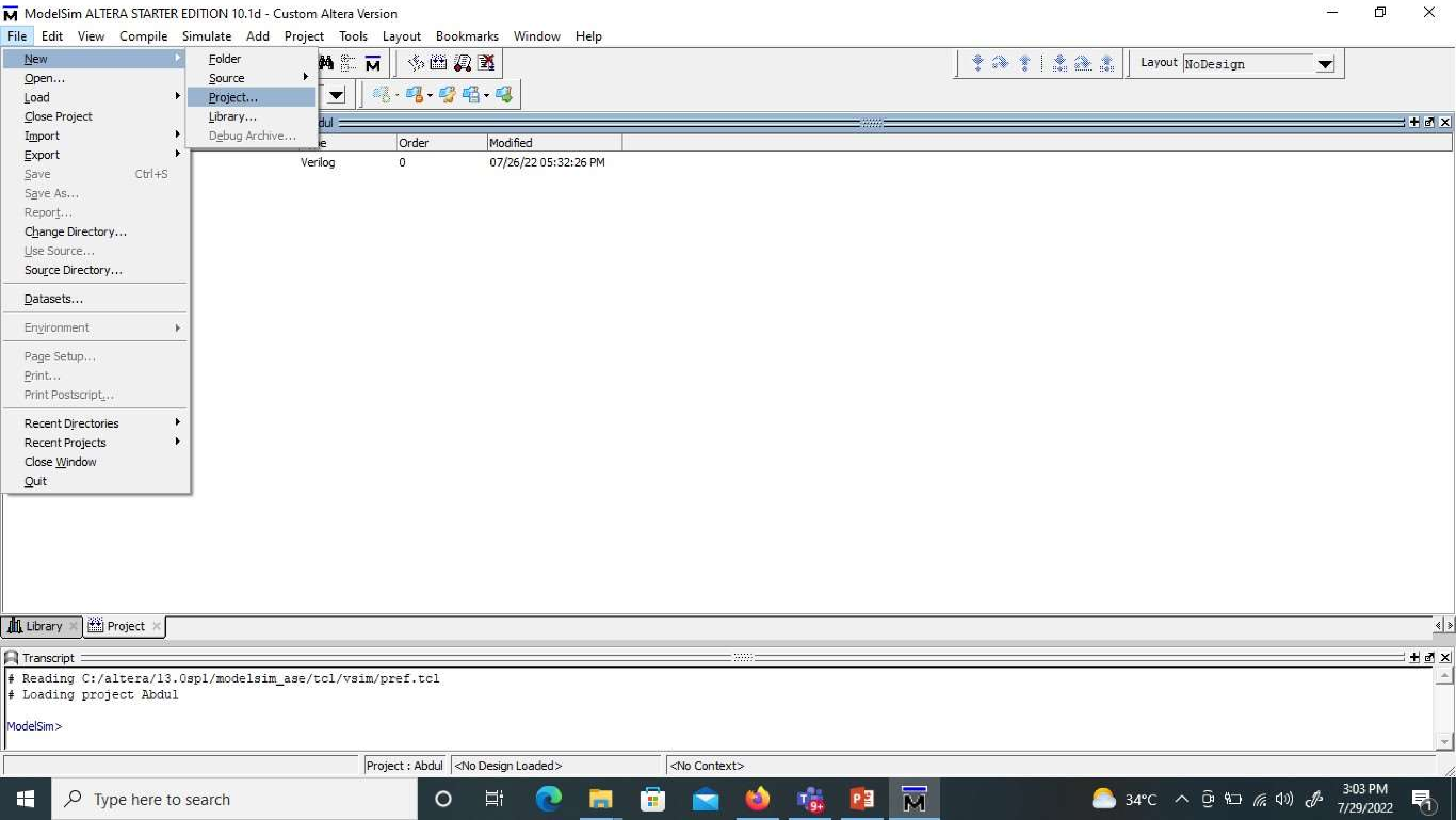


Gate-level modeling

```
module AND_2(output Y, input A, B);
    and(Y, A, B);
endmodule
```

OR

```
module AND_2( Y, A, B);
 Input A,B;
 output Y;
     and(Y, A, B);
endmodule
```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Use Source...
Source Directory...
Datasets...
Environment
Page Setup...
Print...
Print Postscript...
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Source
Project...
Library...
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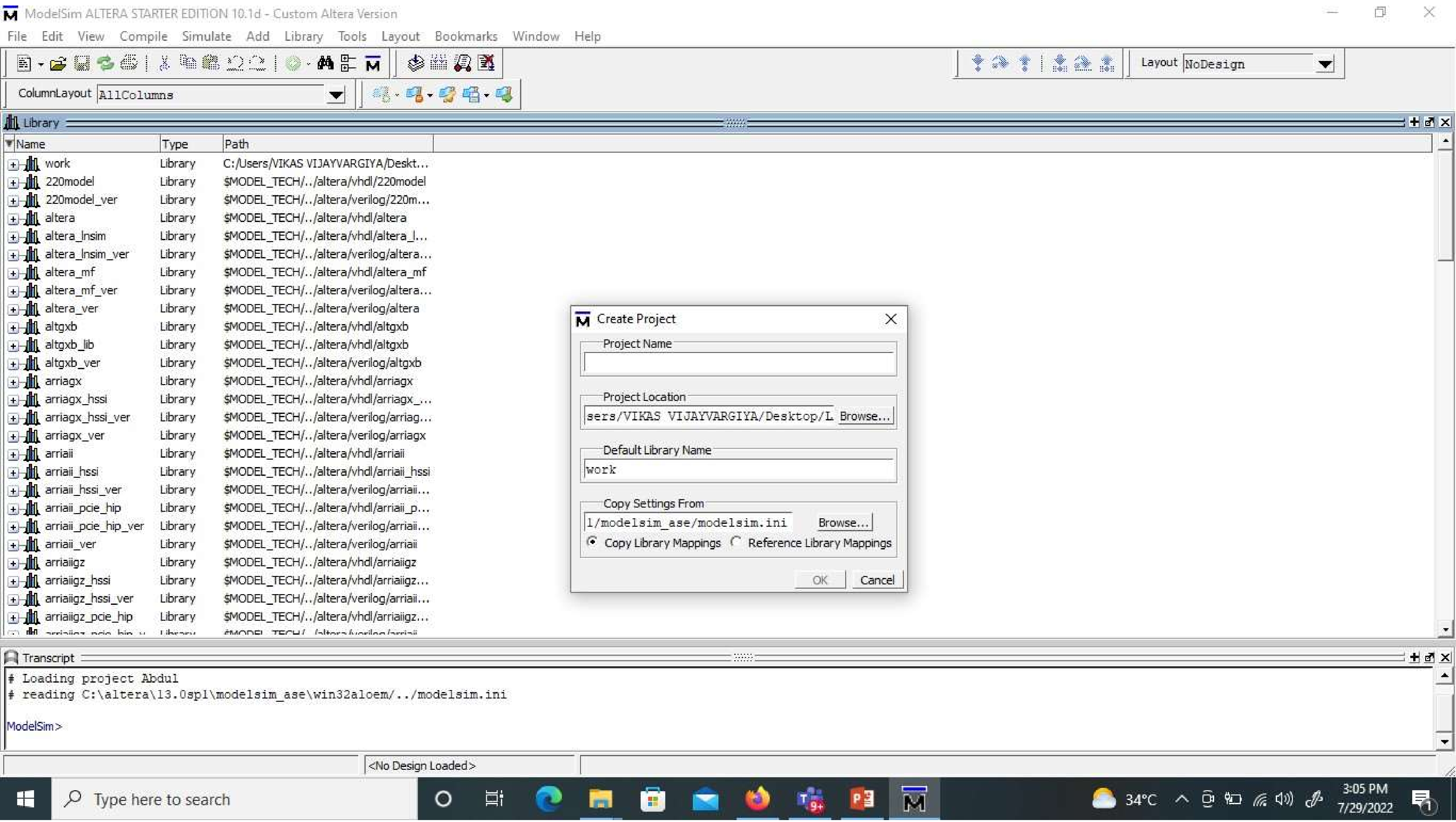

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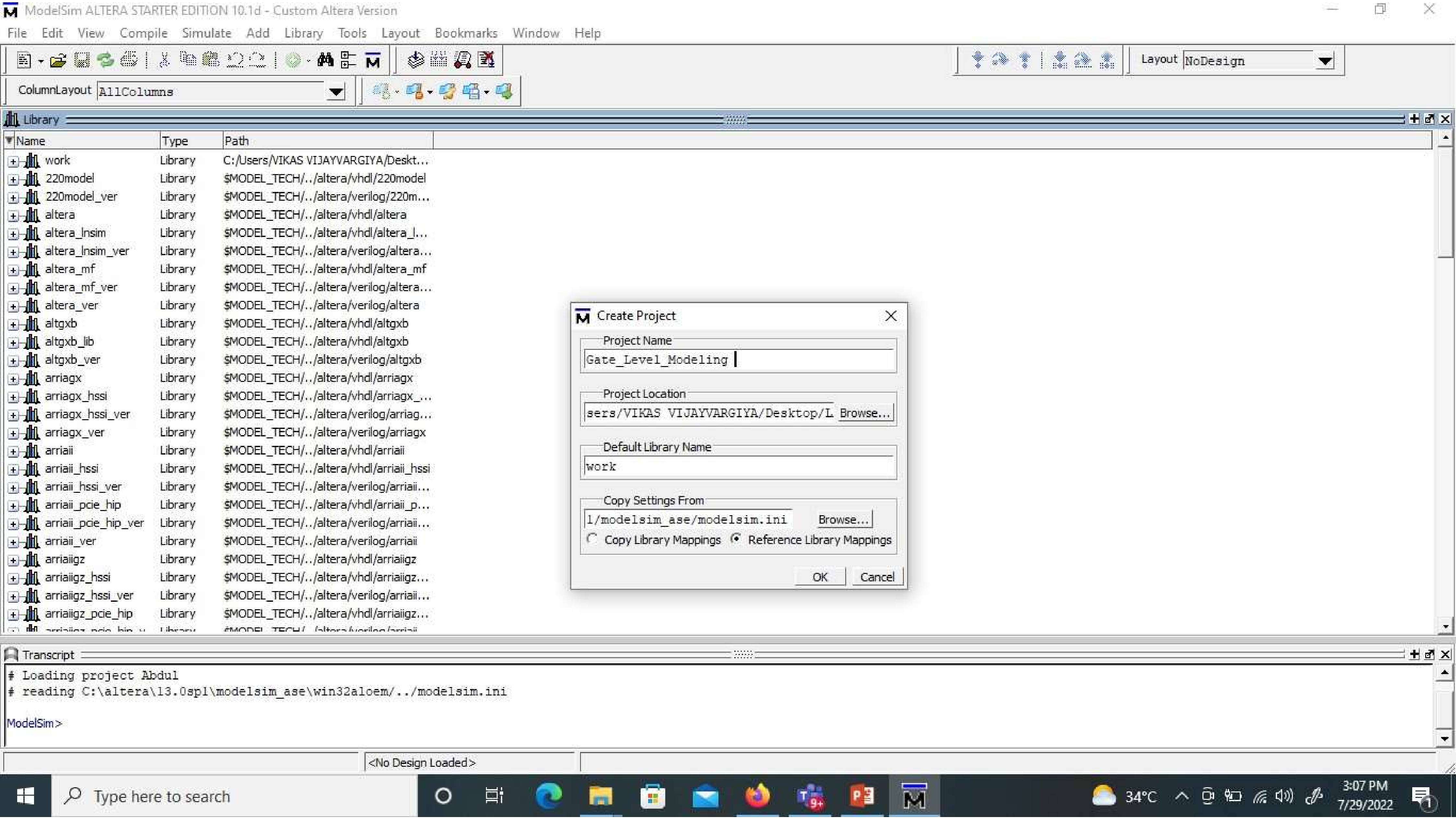
7/29/2022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Create Project
Project Name
Gate_Level_Modeling
Project Location
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Default Library Name
work
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l/modelsim_ase/modelsim.ini Browse...
Copy Library Mappings Reference Library Mappings
OK Cancel
Transcript
Loading project Abdul
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
ModelSim>
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Type here to search
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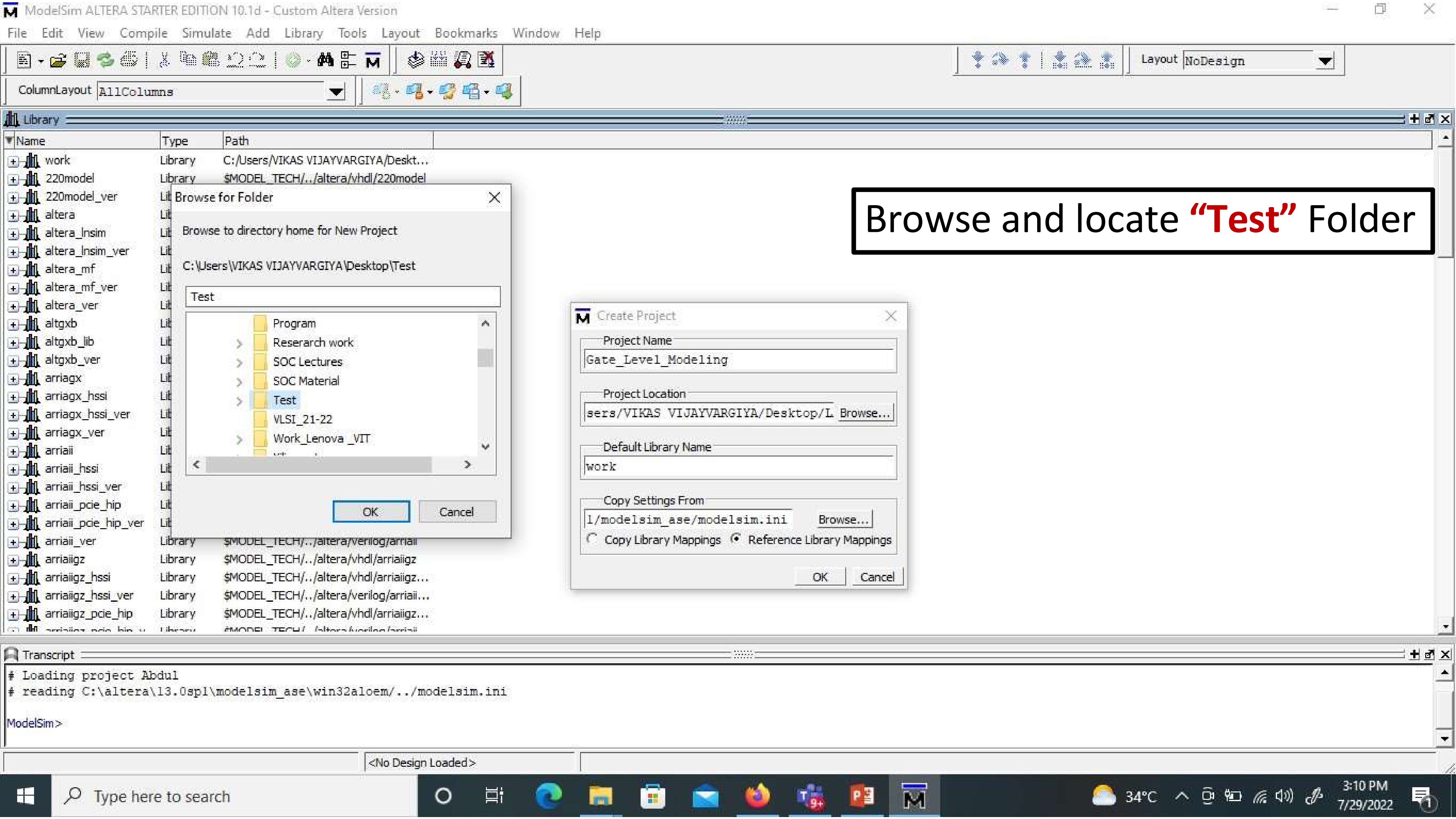
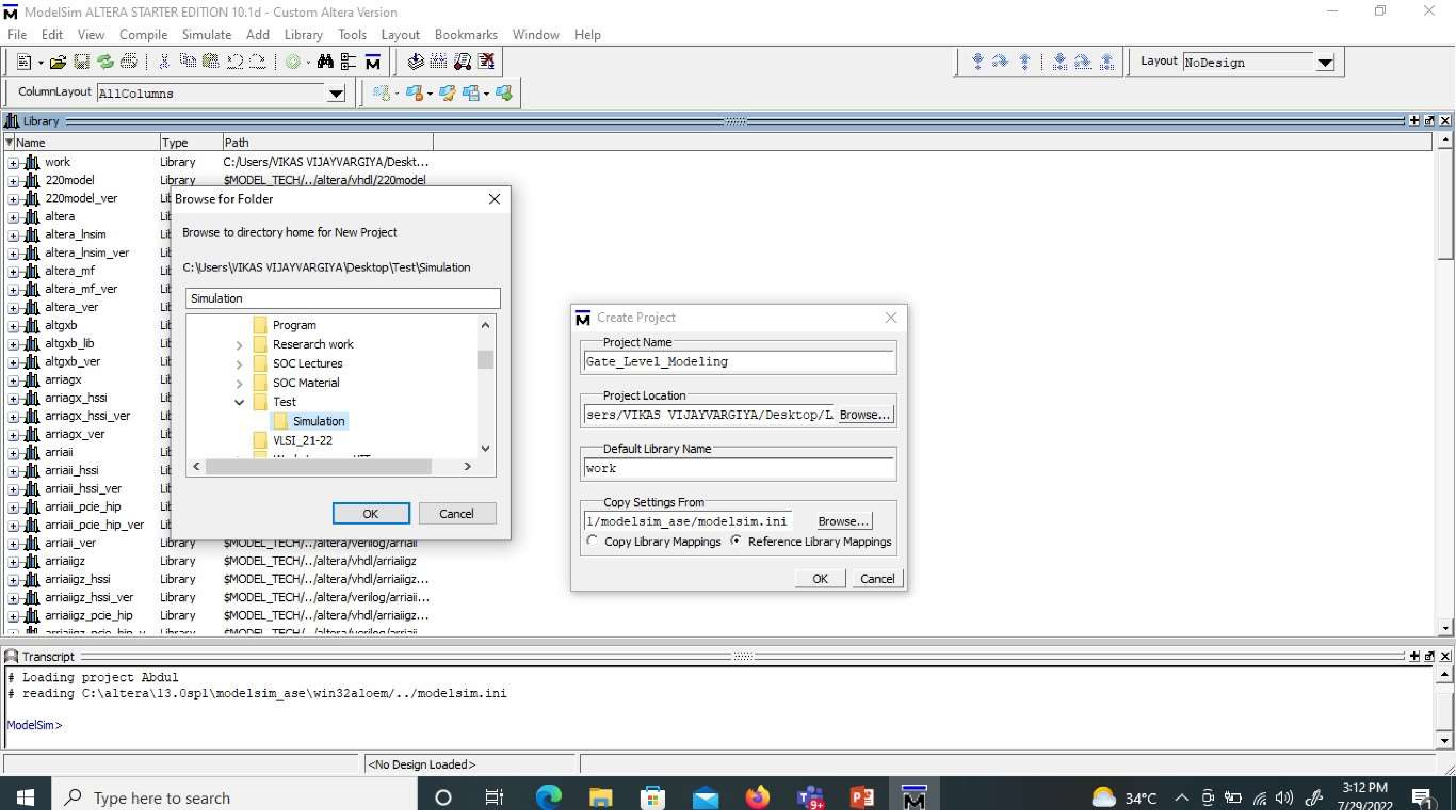
7/29/2022

Create Folder on Desktop: Test

Inside Folder: Simulation folder



Browse and locate **“Test”** Folder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220model Library $MODEL_TECH/../altera/vhdl/220model
220model_ver
altera
altera_lnsim
altera_lnsim_ver
altera_mf
altera_mf_ver
altera_ver
altgxb
altgxb_lib
altgxb_ver
arriagx
arriagx_hssi
arriagx_hssi_ver
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Browse for Folder
Browse to directory home for New Project
C:\Users\VIKAS VIJAYVARGIYA\Desktop\Test\Simulation
Simulation
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SOC Material
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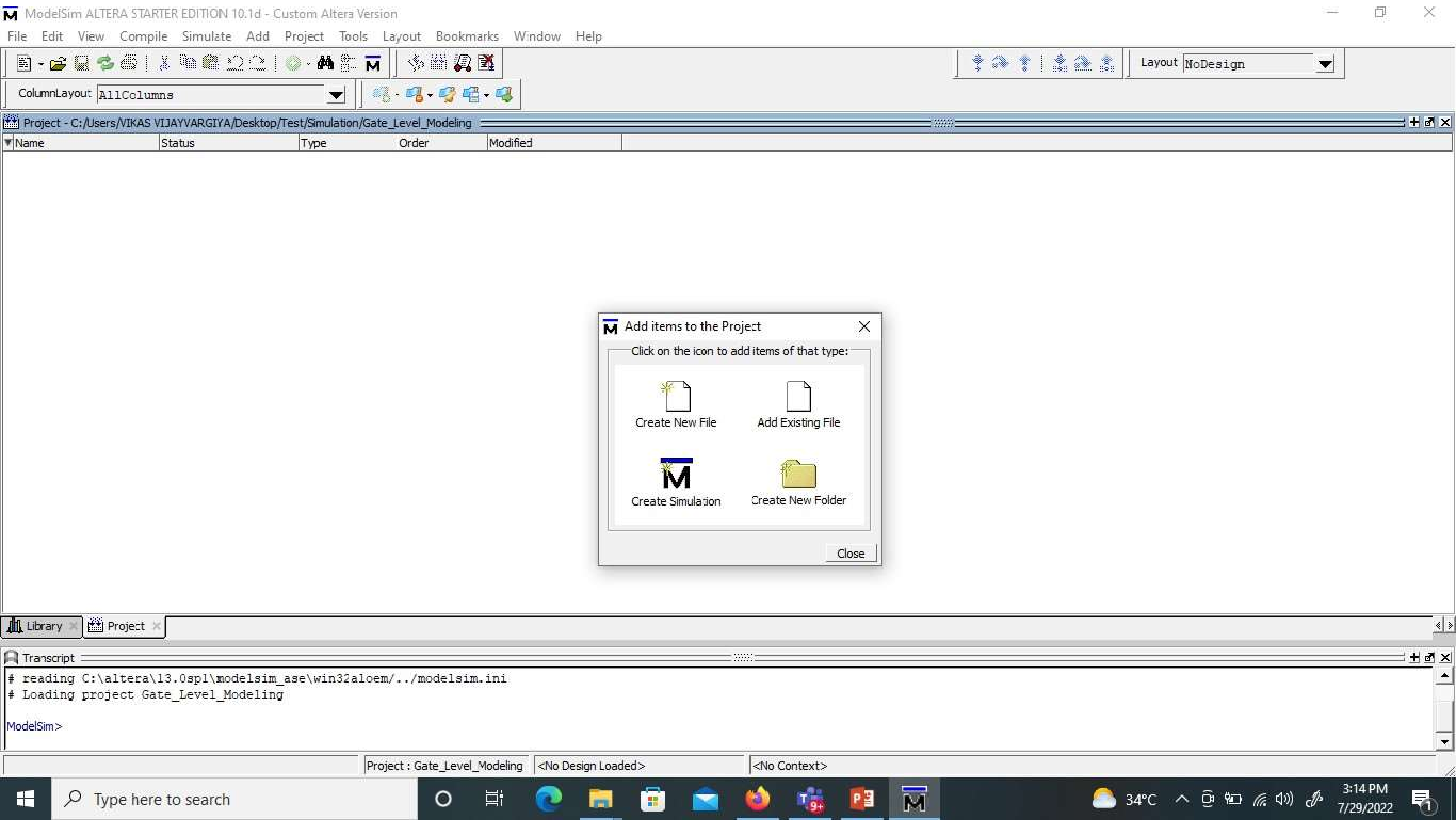
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ColumnLayout AllColumns
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
Add items to the Project
Click on the icon to add items of that type:
Create New File
Add Existing File
Create Simulation
Create New Folder
Close
Library
Project
Transcript
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
Loading project Gate_Level_Modeling
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
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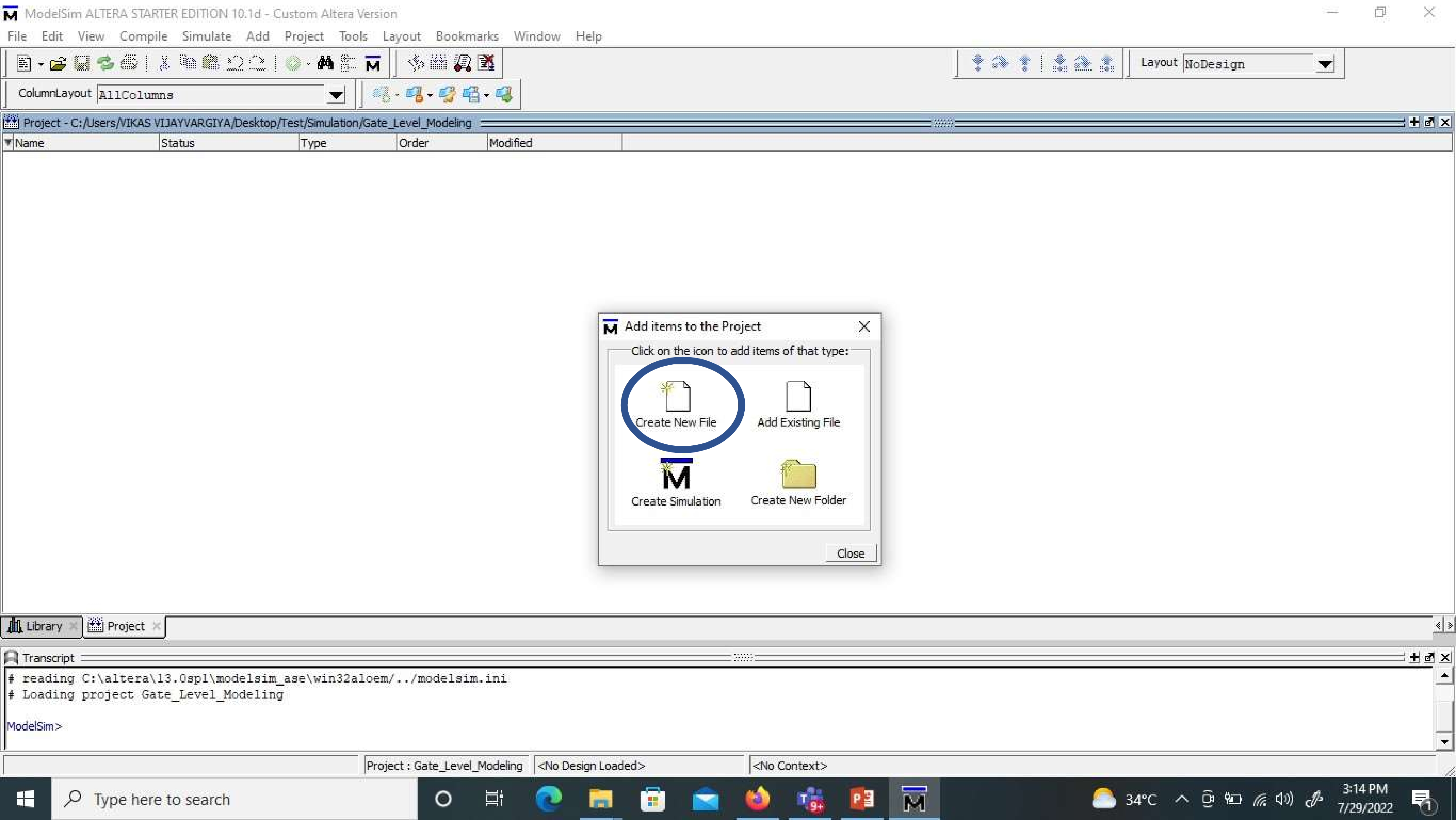

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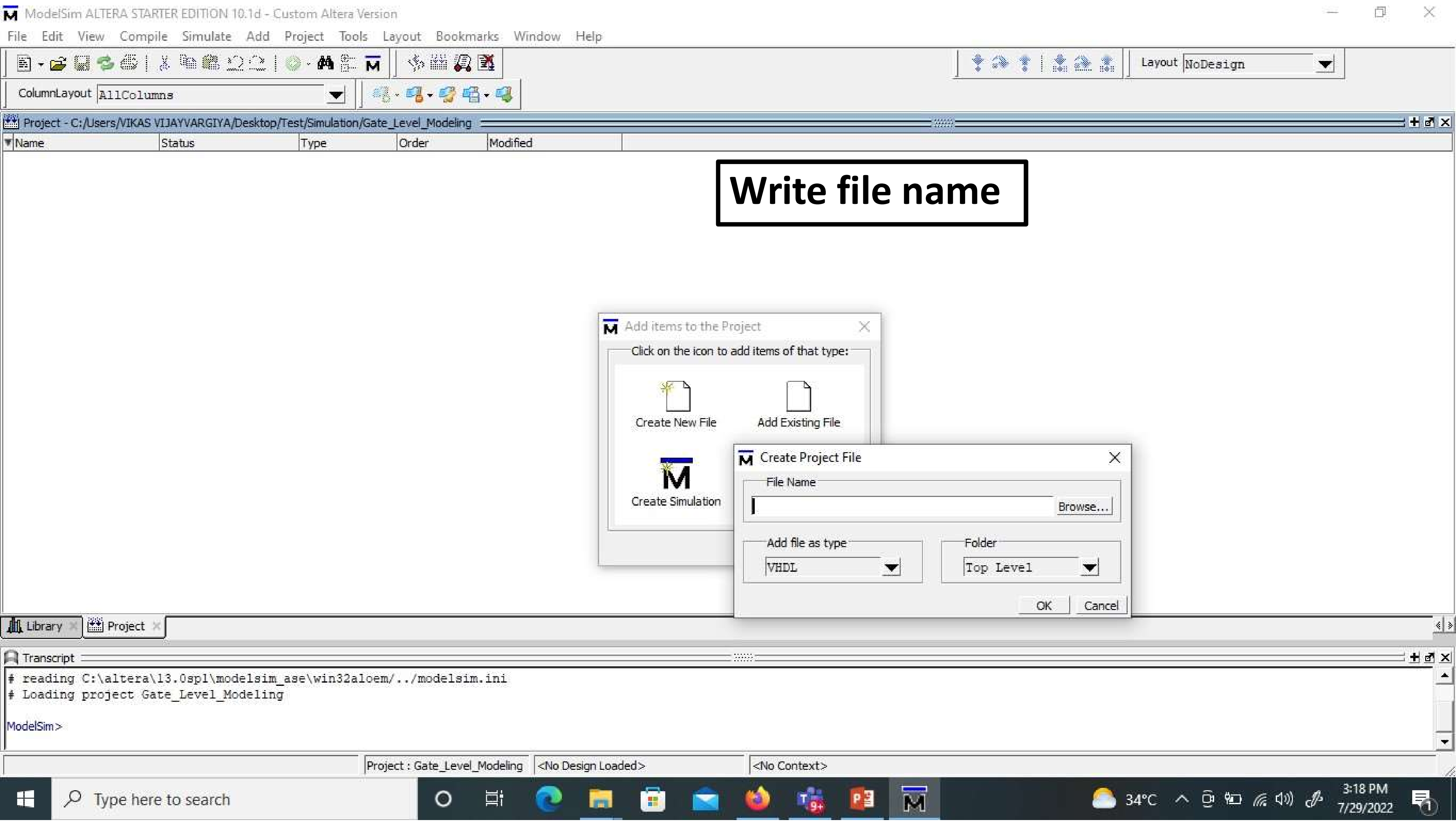
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Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
Write file name
Add items to the Project
Click on the icon to add items of that type:
Create New File
Add Existing File
Create Simulation
Create Project File
File Name
Browse...
Add file as type
VHDL
Folder
Top Level
OK
Cancel
Library
Project
Transcript
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
Loading project Gate_Level_Modeling
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
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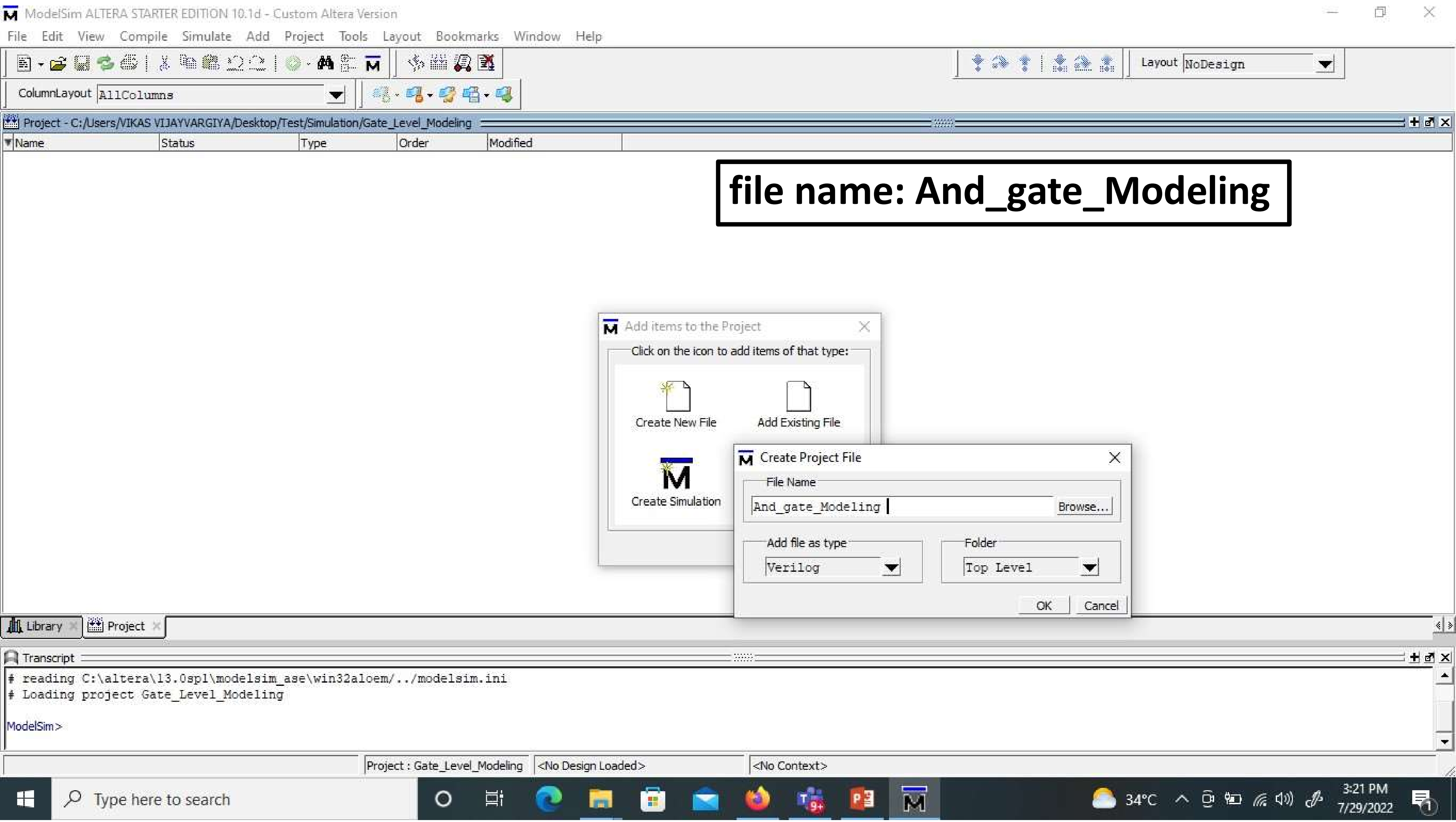
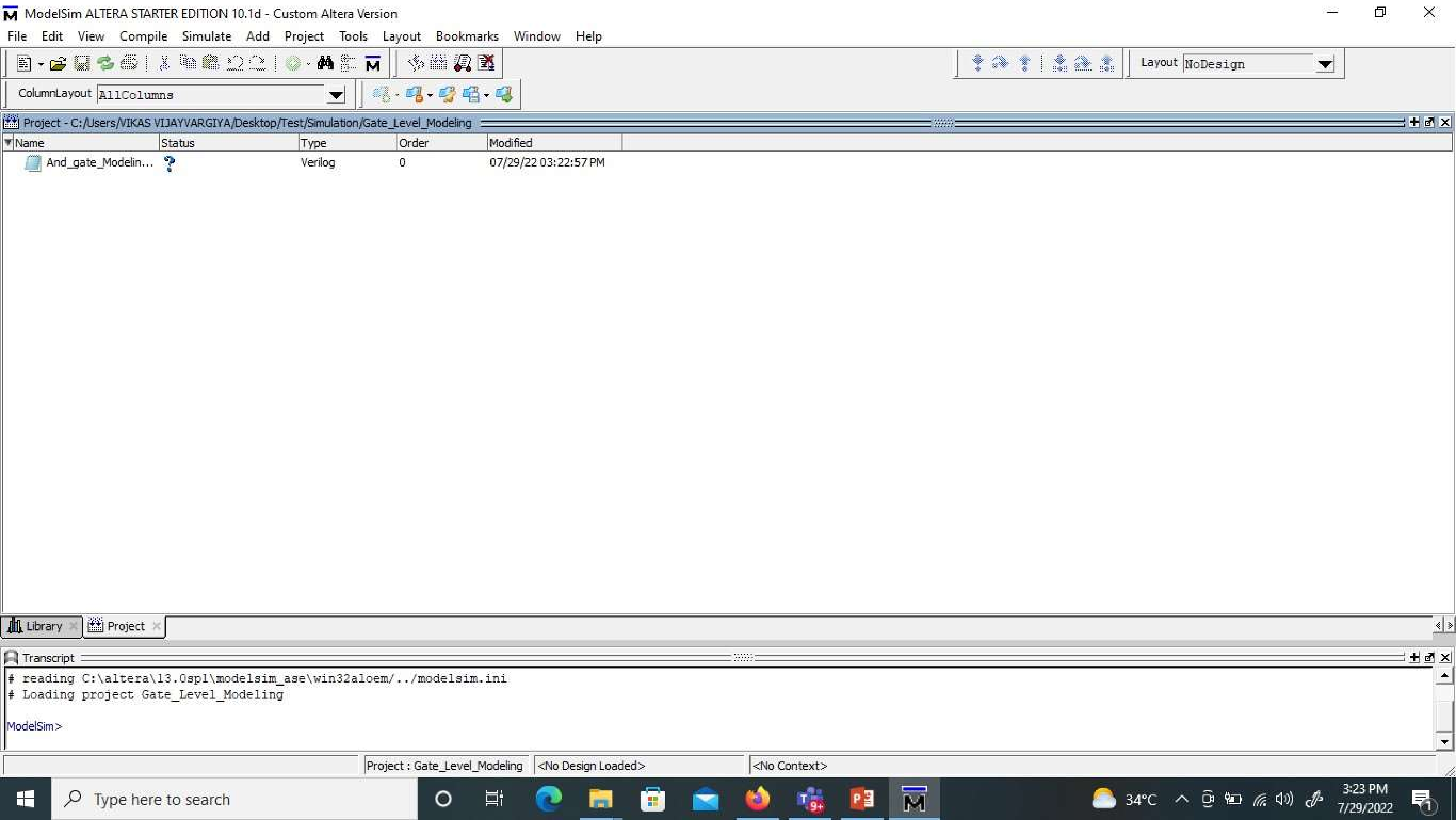
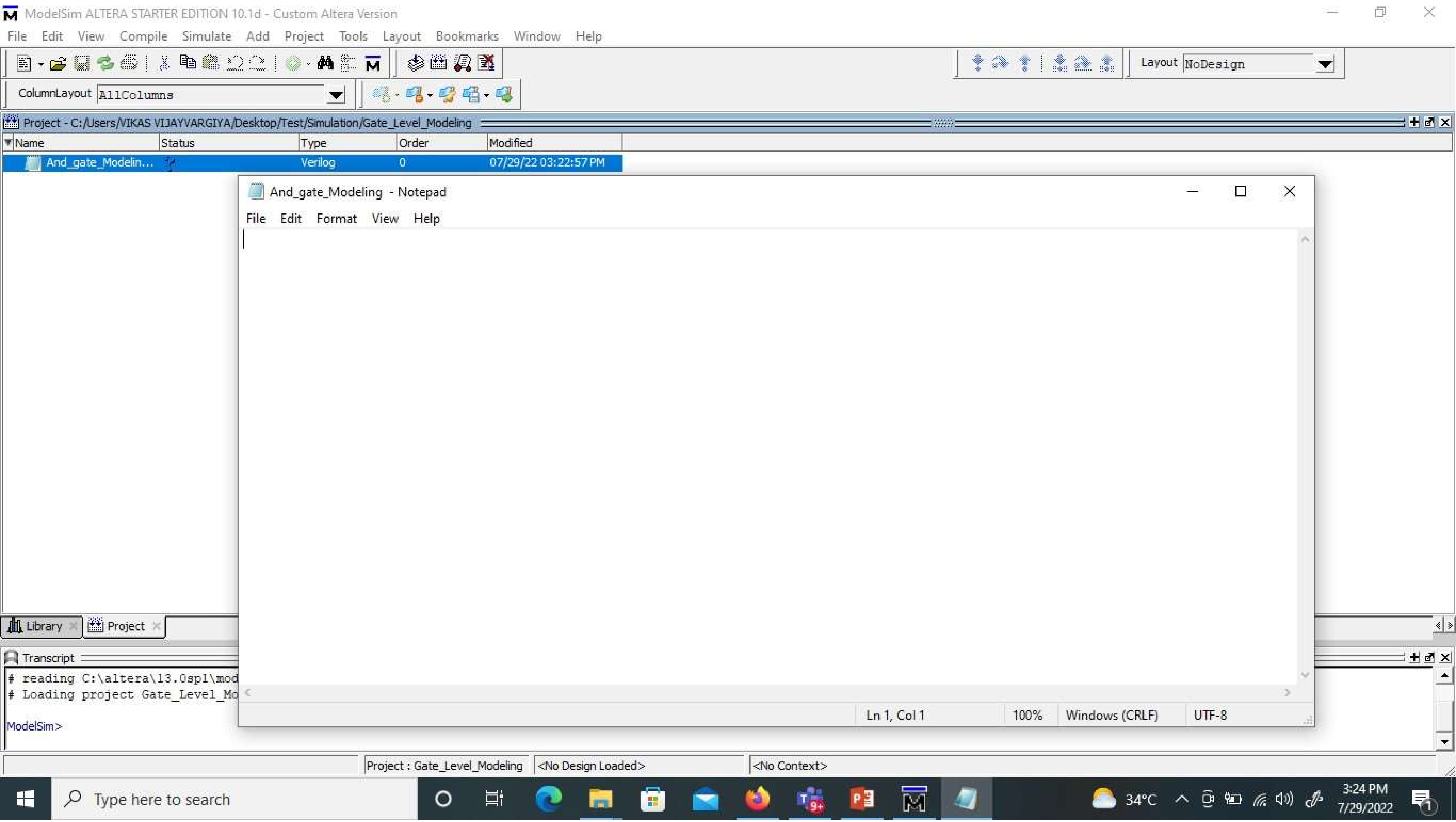
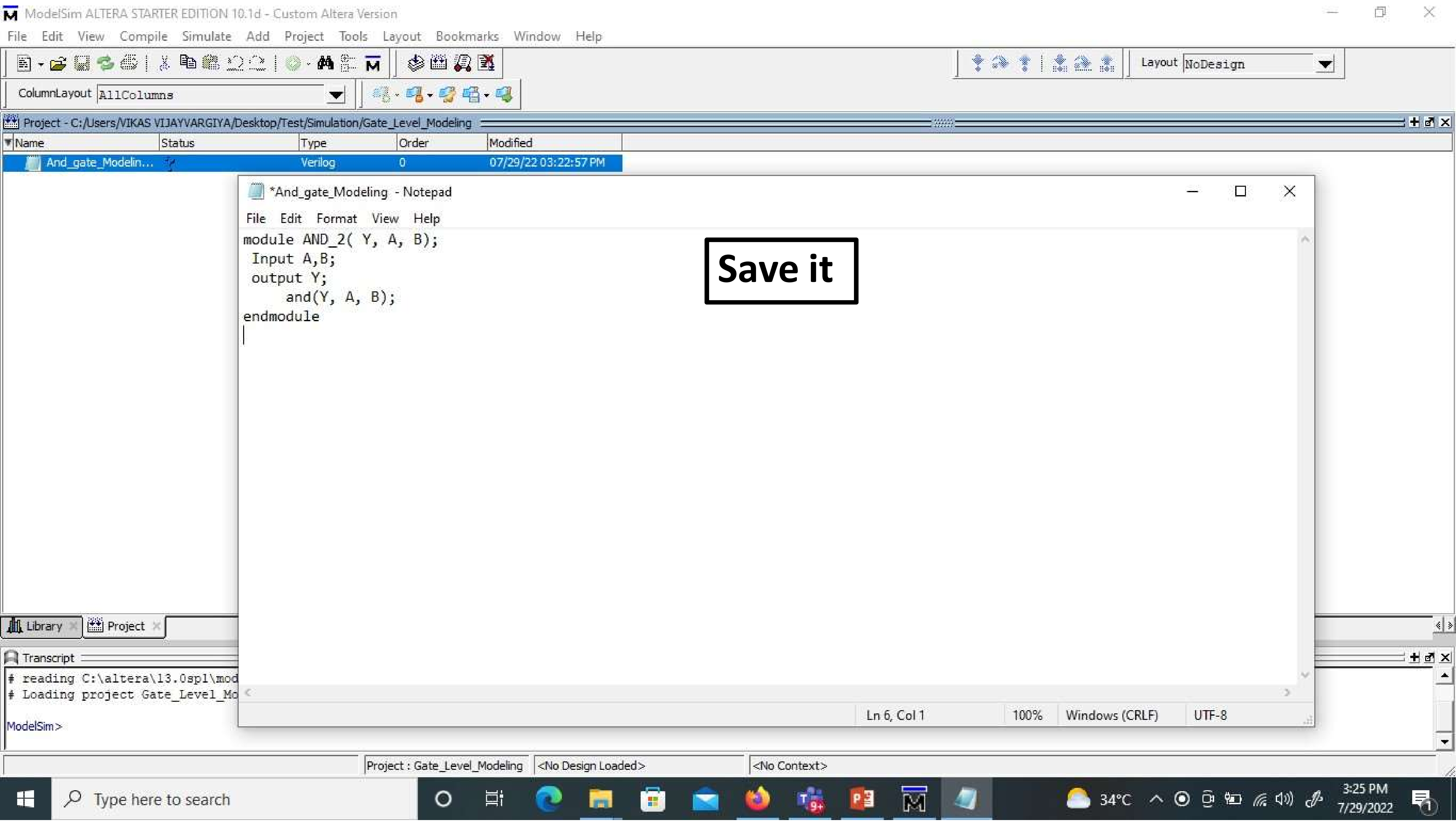
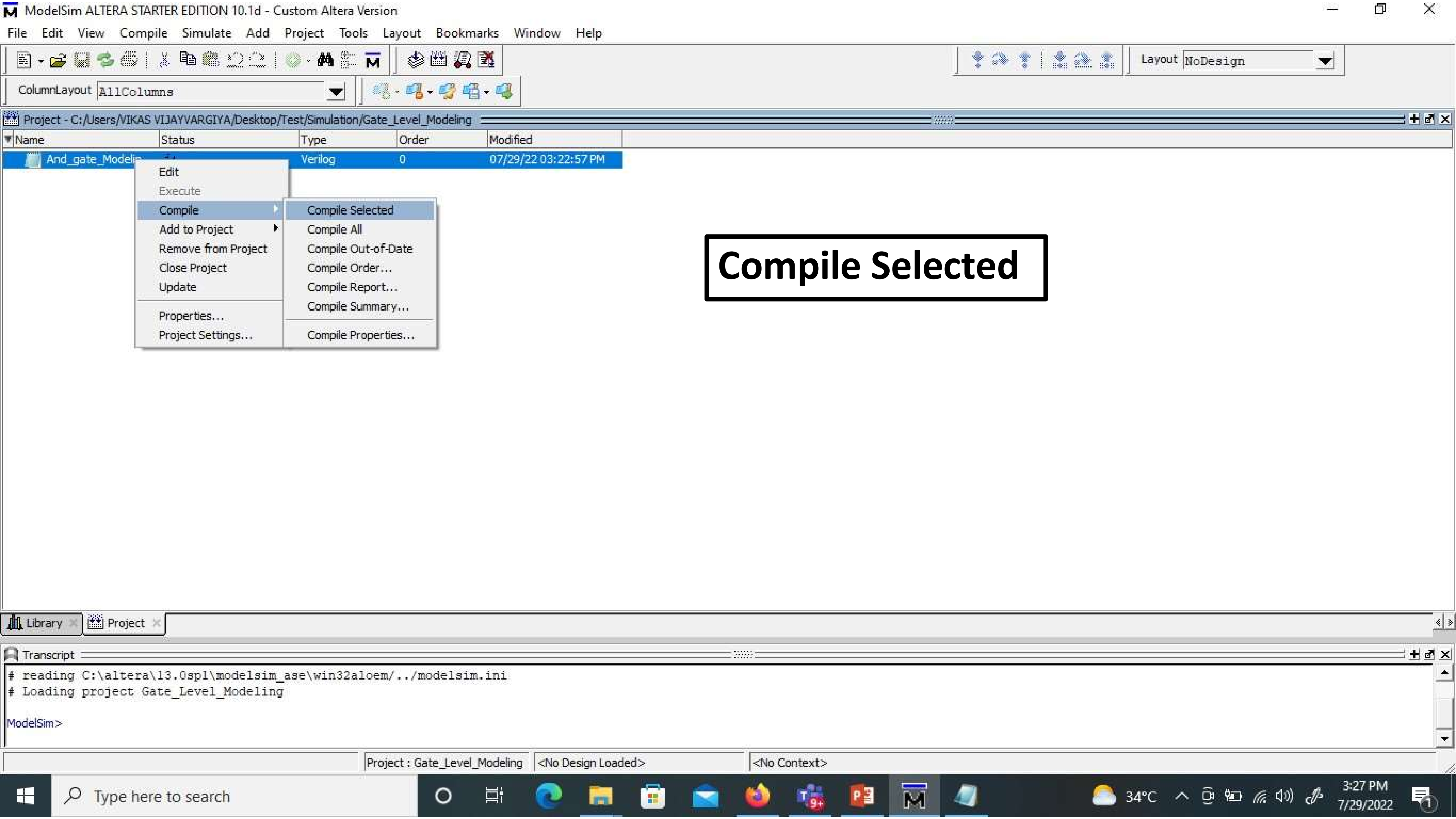
3:18 PM
7/29/2022

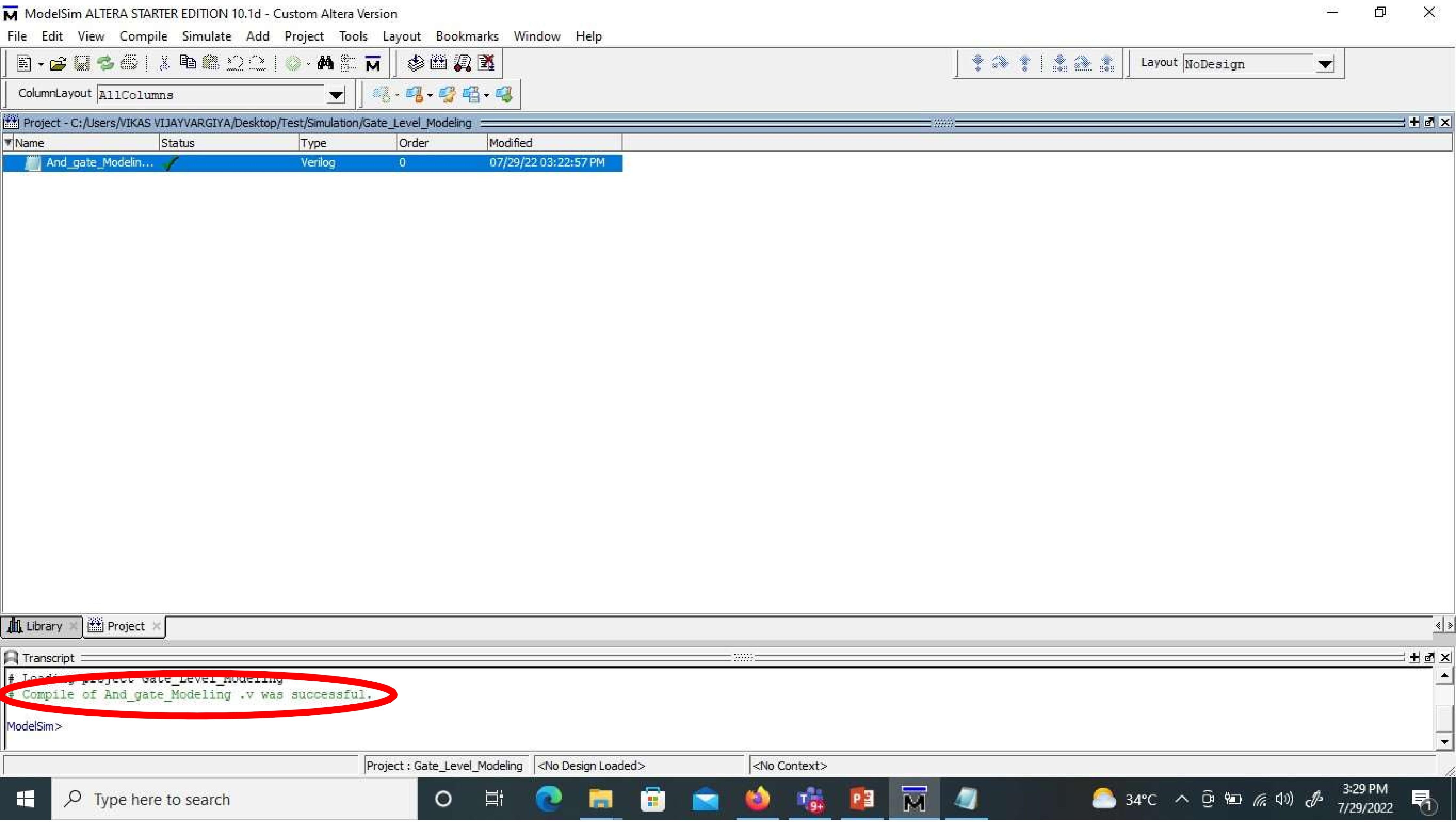

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
file name: And_gate_Modeling
Add items to the Project
Click on the icon to add items of that type:
Create New File
Add Existing File
Create Simulation
Create Project File
File Name
And_gate_Modeling
Browse...
Add file as type
Verilog
Folder
Top Level
OK
Cancel
Transcript
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
Loading project Gate_Level_Modeling
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
<No Context>

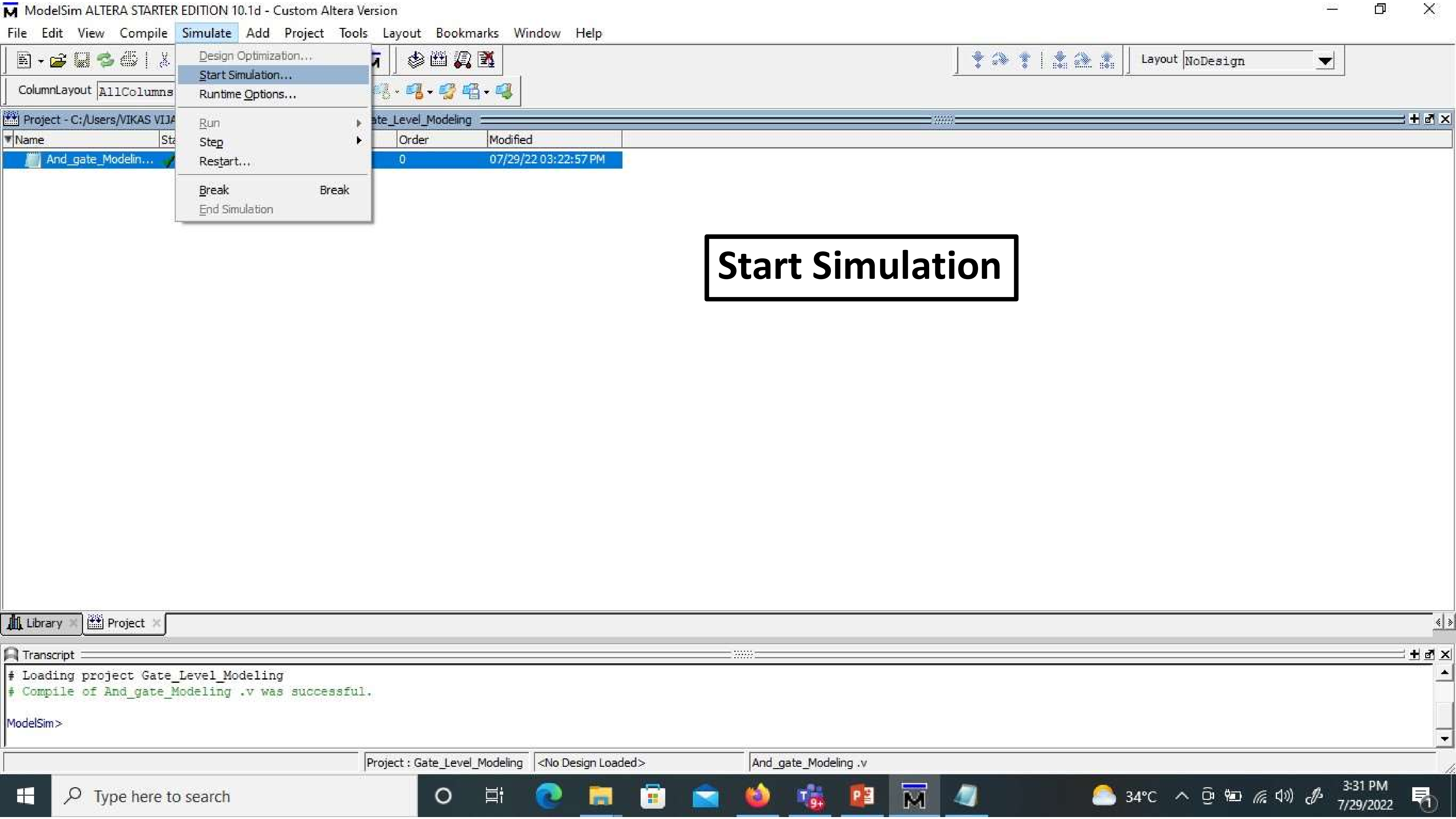

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Layout NoDesign
ColumnLayout AllColumns
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
And_gate_Modelin... ? Verilog 0 07/29/22 03:22:57 PM
Library Project
Transcript
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
Loading project Gate_Level_Modeling
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
<No Context>
Type here to search
34°C
3:23 PM
7/29/2022


ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
ColumnLayout AllColumns
Layout NoDesign
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
And_gate_Modelin... Verilog 0 07/29/22 03:22:57 PM
And_gate_Modeling - Notepad
File Edit Format View Help
Ln 1, Col 1 100% Windows (CRLF) UTF-8
Library Project
Transcript
reading C:\altera\13.0sp1\mod
Loading project Gate_Level_Mo
ModelSim>
Project : Gate_Level_Modeling <No Design Loaded> <No Context>
Type here to search
34°C
3:24 PM
7/29/2022

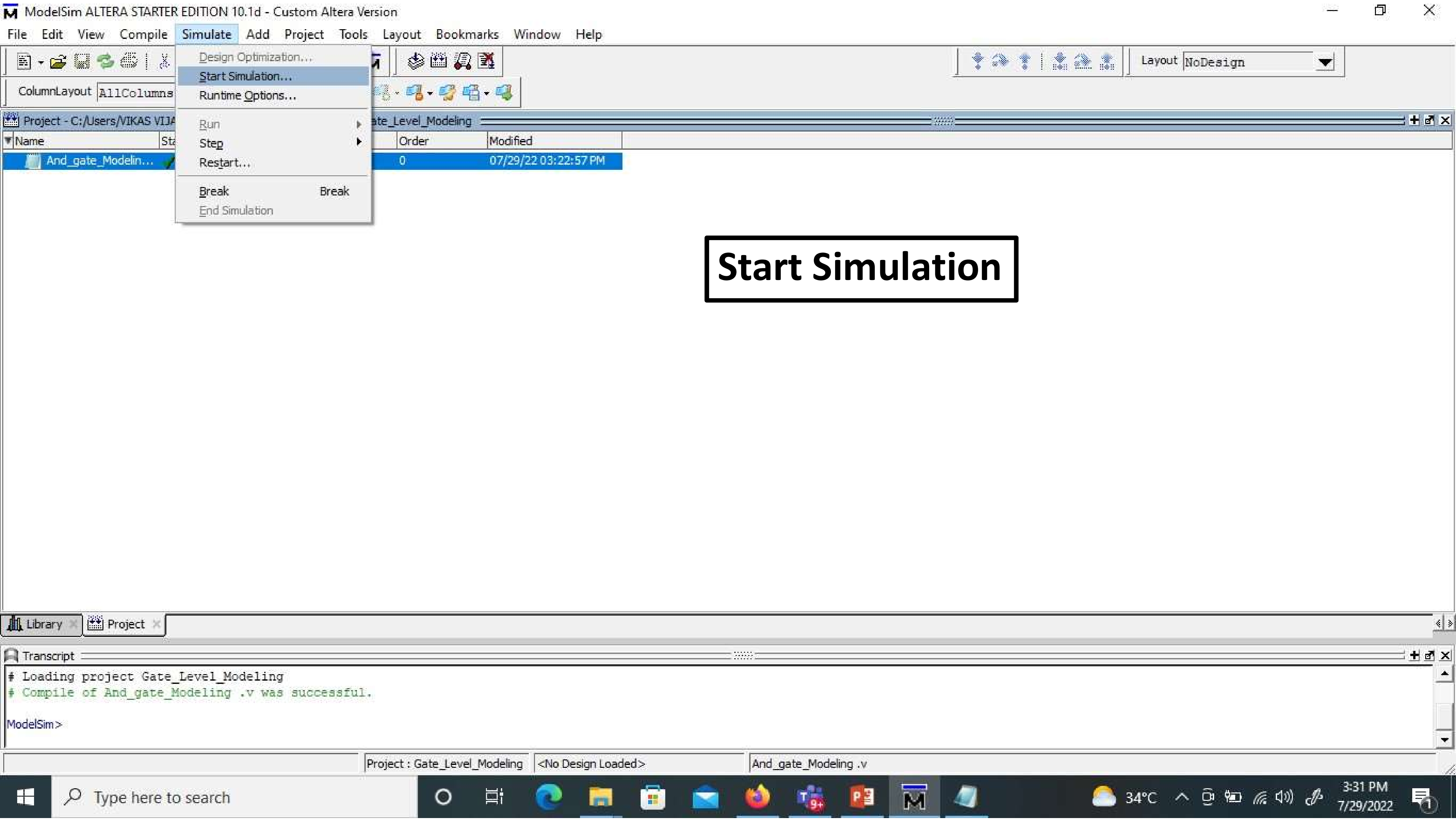

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Layout NoDesign
ColumnLayout AllColumns
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
And_gate_Modelin... Verilog 0 07/29/22 03:22:57 PM
*And_gate_Modeling - Notepad
File Edit Format View Help
module AND_2(Y, A, B);
Input A,B;
output Y;
and(Y, A, B);
endmodule
Save it
Ln 6, Col 1 100% Windows (CRLF) UTF-8
Library Project
Transcript
reading C:\altera\13.0sp1\mod
Loading project Gate_Level_Mo
ModelSim>
Project : Gate_Level_Modeling <No Design Loaded> <No Context>
Type here to search
34°C 3:25 PM 7/29/2022

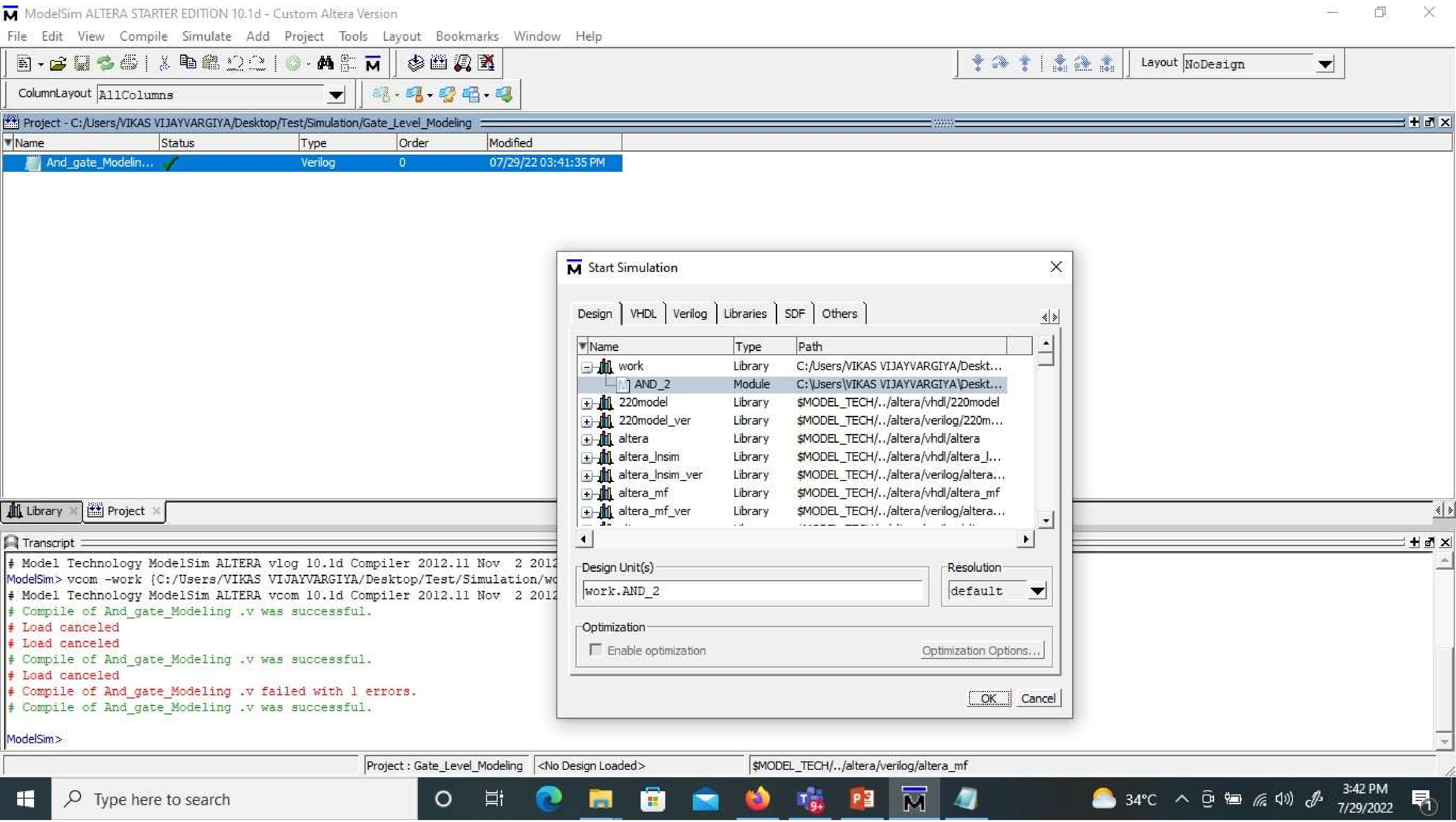

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Layout NoDesign
ColumnLayout AllColumns
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
And_gate_Modelin Verilog 0 07/29/22 03:22:57 PM
Edit
Execute
Compile
Add to Project
Remove from Project
Close Project
Update
Properties...
Project Settings...
Compile Selected
Compile All
Compile Out-of-Date
Compile Order...
Compile Report...
Compile Summary...
Compile Properties...
Compile Selected
Library
Project
Transcript
reading C:\altera\13.0sp1\modelsim_ase\win32aloem/../modelsim.ini
Loading project Gate_Level_Modeling
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
<No Context>
Type here to search
34°C
3:27 PM
7/29/2022

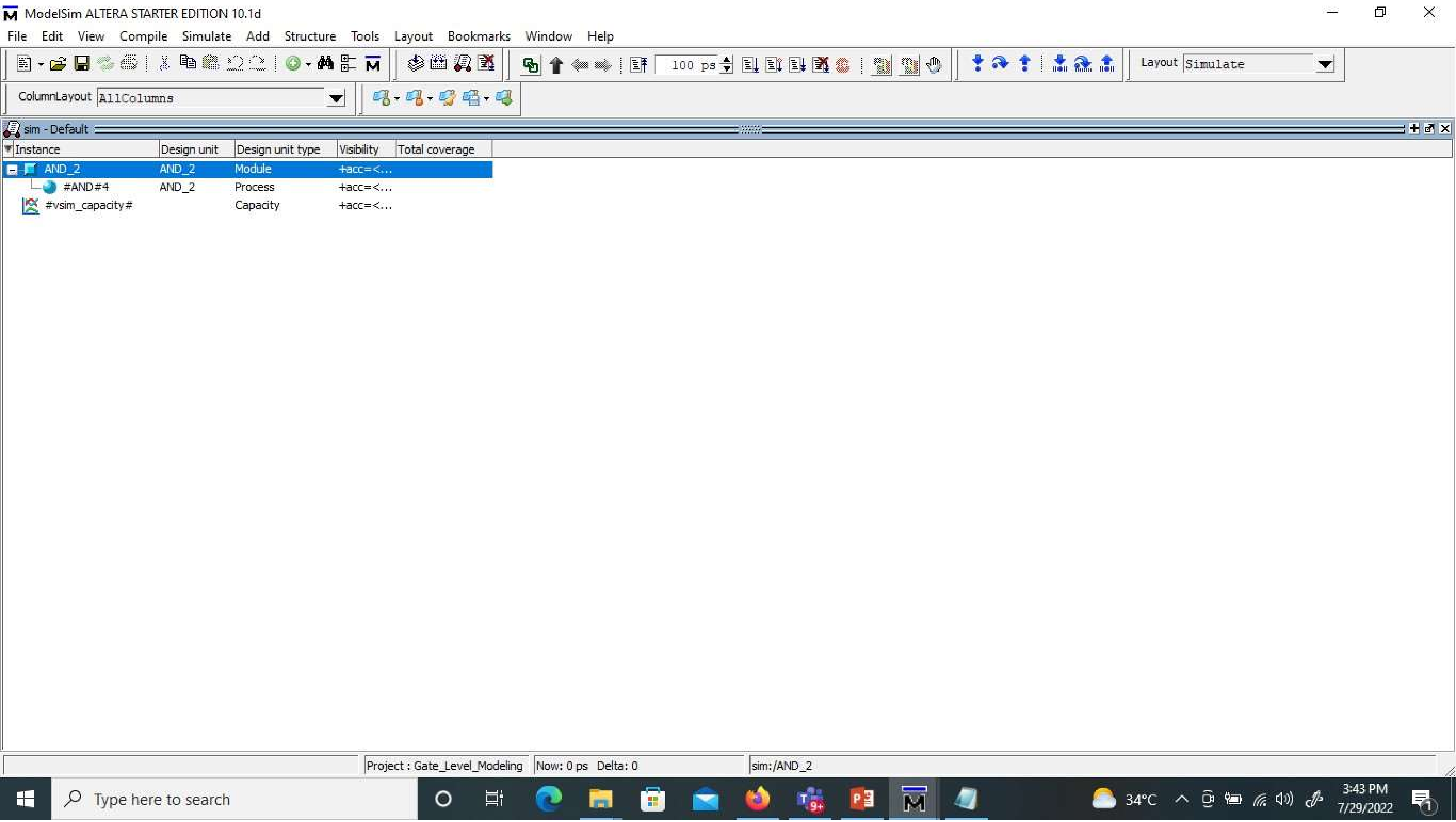

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Layout NoDesign
ColumnLayout AllColumns
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
Name Status Type Order Modified
And_gate_Modelin... Verilog 0 07/29/22 03:22:57 PM
Library
Project
Transcript
Compile of And_gate_Modeling .v was successful.
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
<No Context>
Type here to search
34°C
3:29 PM
7/29/2022

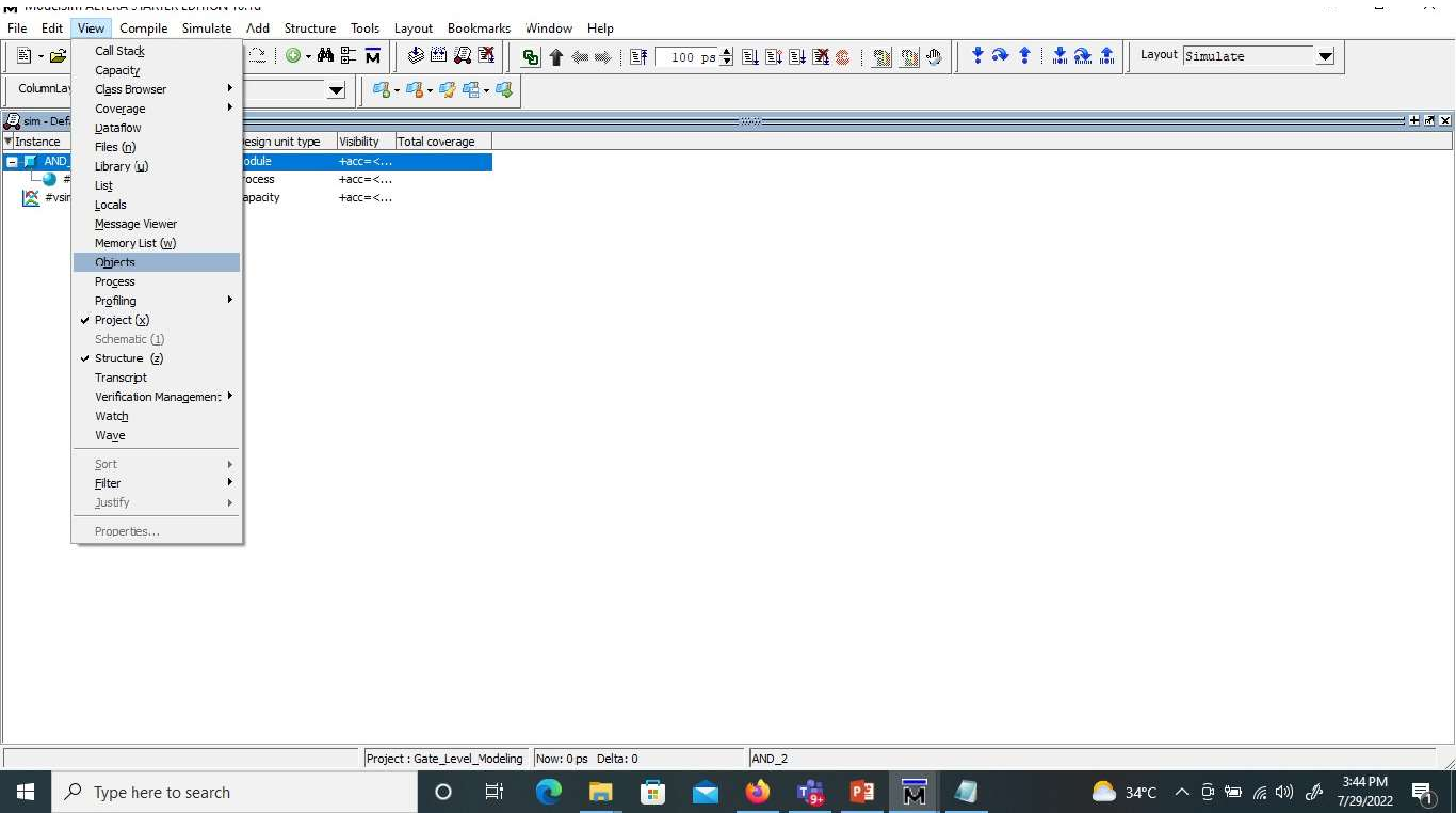

ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Design Optimization...
Start Simulation...
Runtime Options...
Run
Step
Restart...
Break Break
End Simulation
ColumnLayout AllColumns
Layout NoDesign
Name
Order
Modified
And_gate_Modelin...
0
07/29/22 03:22:57 PM
Start Simulation
Library
Project
Transcript
Loading project Gate_Level_Modeling
Compile of And_gate_Modeling .v was successful.
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>
And_gate_Modeling .v
Type here to search
34°C
3:31 PM
7/29/2022

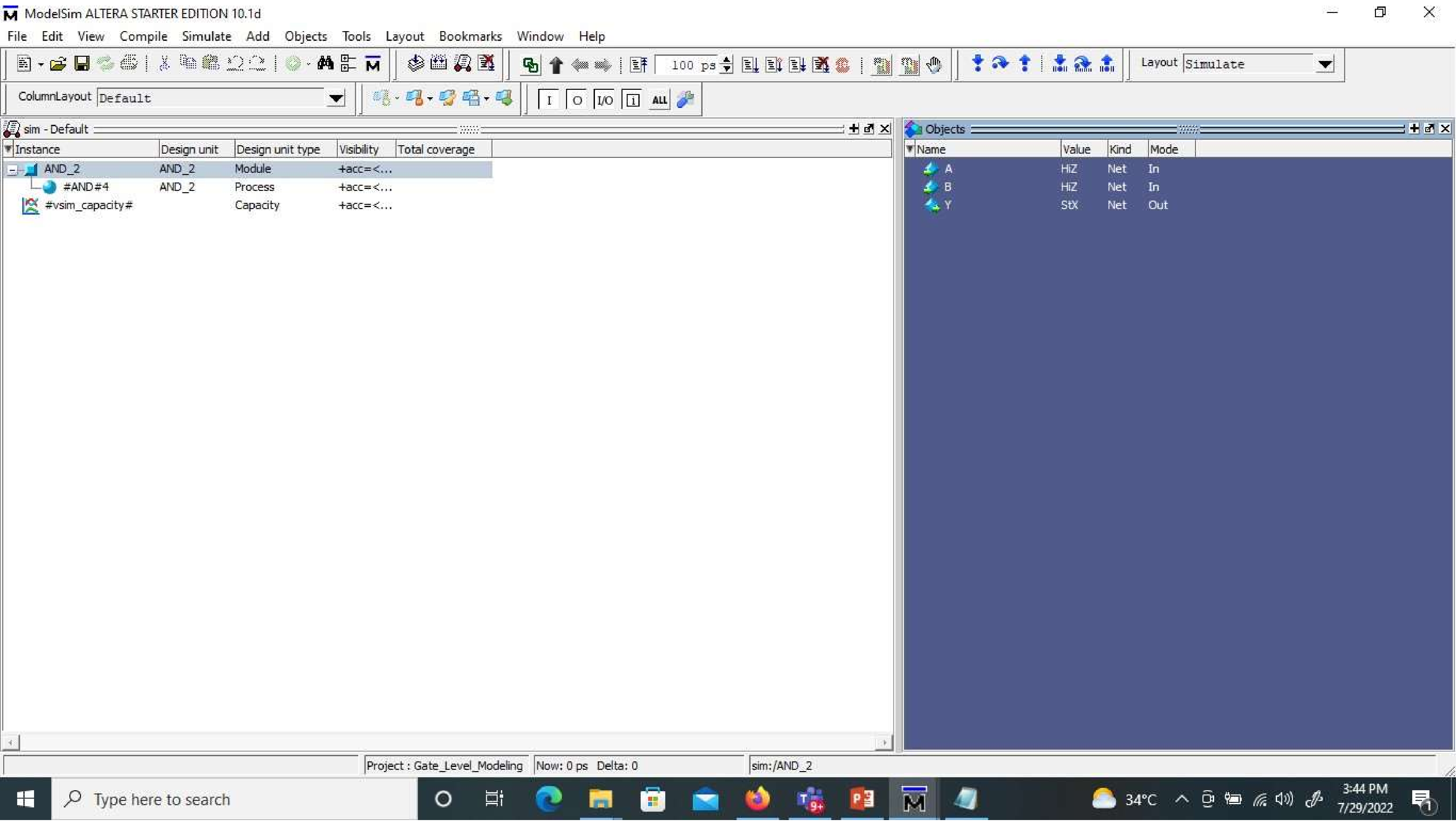
Start Simulation

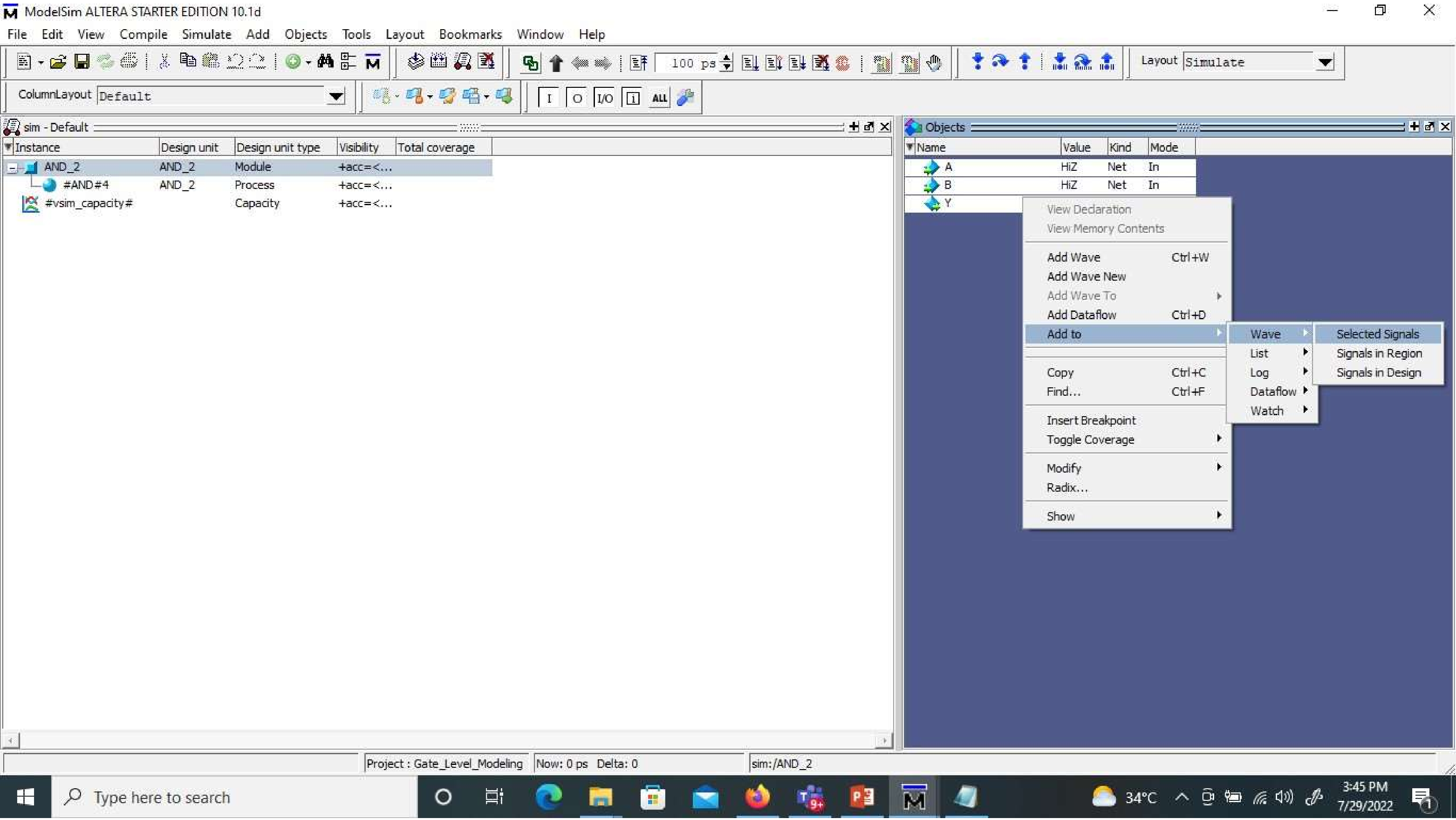



ModelSim ALTERA STARTER EDITION 10.1d - Custom Altera Version
File Edit View Compile Simulate Add Project Tools Layout Bookmarks Window Help
Project - C:/Users/VIKAS VIJAYVARGIYA/Desktop/Test/Simulation/Gate_Level_Modeling
And_gate_Modelin... Verilog 0 07/29/22 03:41:35 PM
Start Simulation
Design VHDL Verilog Libraries SDF Others
work Library
AND_2 Module
Design Unit(s)
work.AND_2
Resolution
default
Optimization
Enable optimization
Optimization Options...
OK Cancel
Transcript
Compile of And_gate_Modeling .v was successful.
Load canceled
Load canceled
Compile of And_gate_Modeling .v was successful.
Load canceled
Compile of And_gate_Modeling .v failed with 1 errors.
Compile of And_gate_Modeling .v was successful.
ModelSim>
Project : Gate_Level_Modeling
<No Design Loaded>

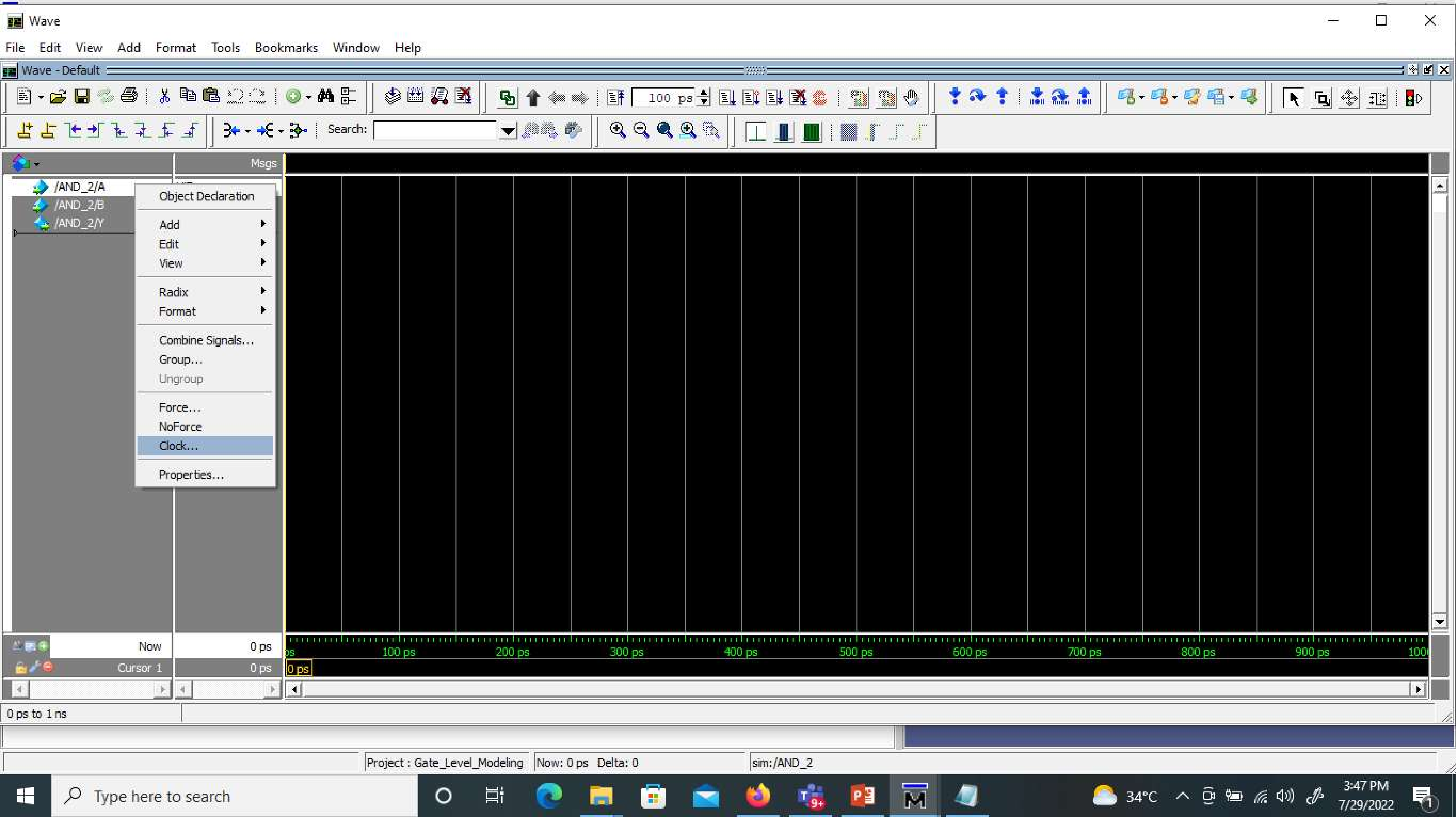

ModelSim ALTERA STARTER EDITION 10.1d
File Edit View Compile Simulate Add Structure Tools Layout Bookmarks Window Help
100 ps
Layout Simulate
ColumnLayout AllColumns
sim - Default
Instance
Design unit
Design unit type
Visibility
Total coverage
AND_2 AND_2 Module +acc=<...
#AND#4 AND_2 Process +acc=<...
#vsim_capacity# Capacity +acc=<...
Project : Gate_Level_Modeling
Now: 0 ps Delta: 0
sim:/AND_2
Type here to search
34°C
3:43 PM
7/29/2022

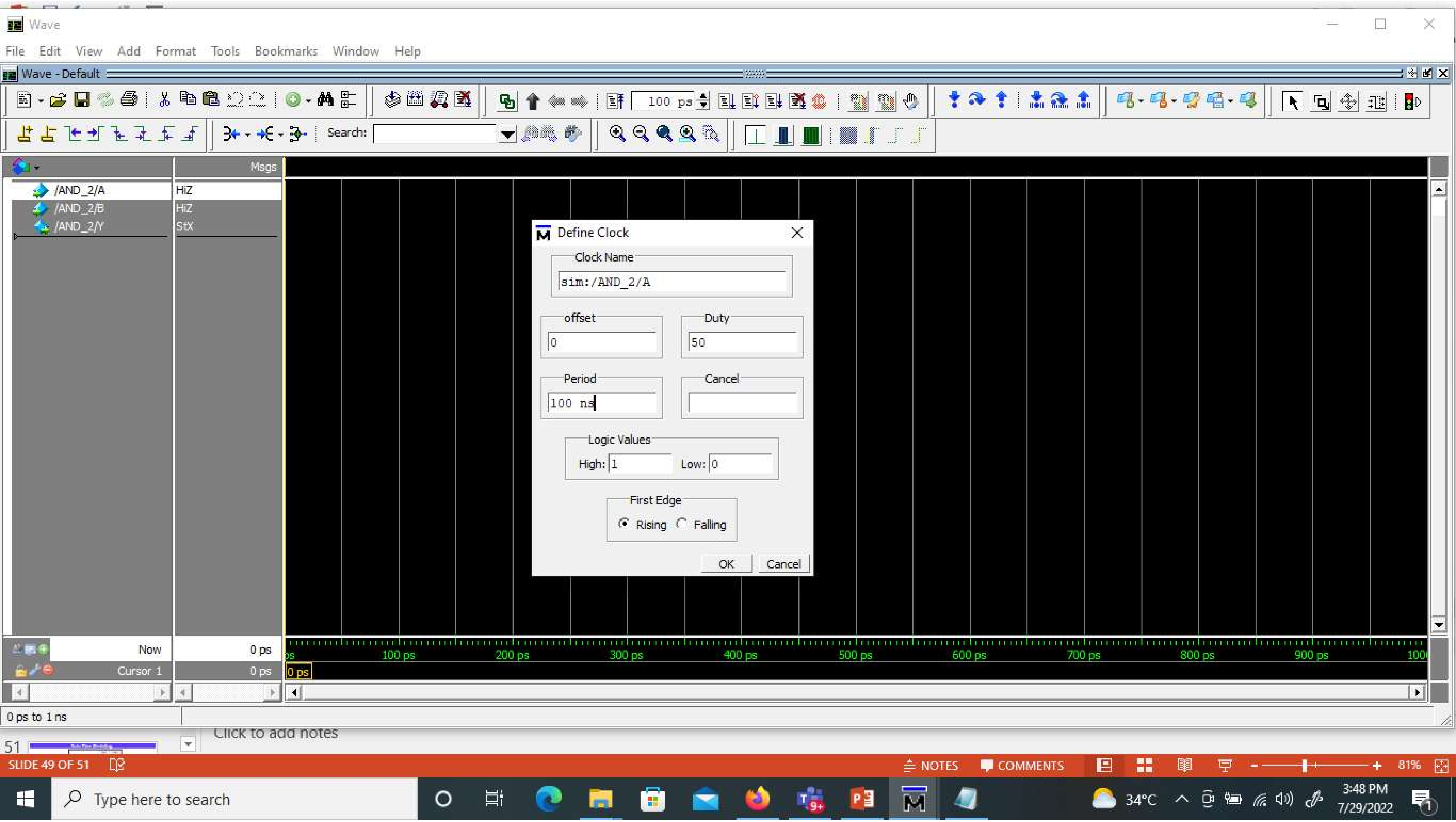

File Edit View Compile Simulate Add Structure Tools Layout Bookmarks Window Help
100 ps
Layout Simulate
ColumnLay
sim - Def
Instance
esign unit type
Visibility
Total coverage
AND_
odule +acc=<...
rocess +acc=<...
apacity +acc=<...
#vsir
Call Stack
Capacity
Class Browser
Coverage
Dataflow
Files (n)
Library (u)
List
Locals
Message Viewer
Memory List (w)
Objects
Process
Profiling
Project (x)
Schematic (1)
Structure (z)
Transcript
Verification Management
Watch
Wave
Sort
Filter
Justify
Properties...
Project : Gate_Level_Modeling
Now: 0 ps Delta: 0
AND_2
Type here to search
34°C
3:44 PM
7/29/2022

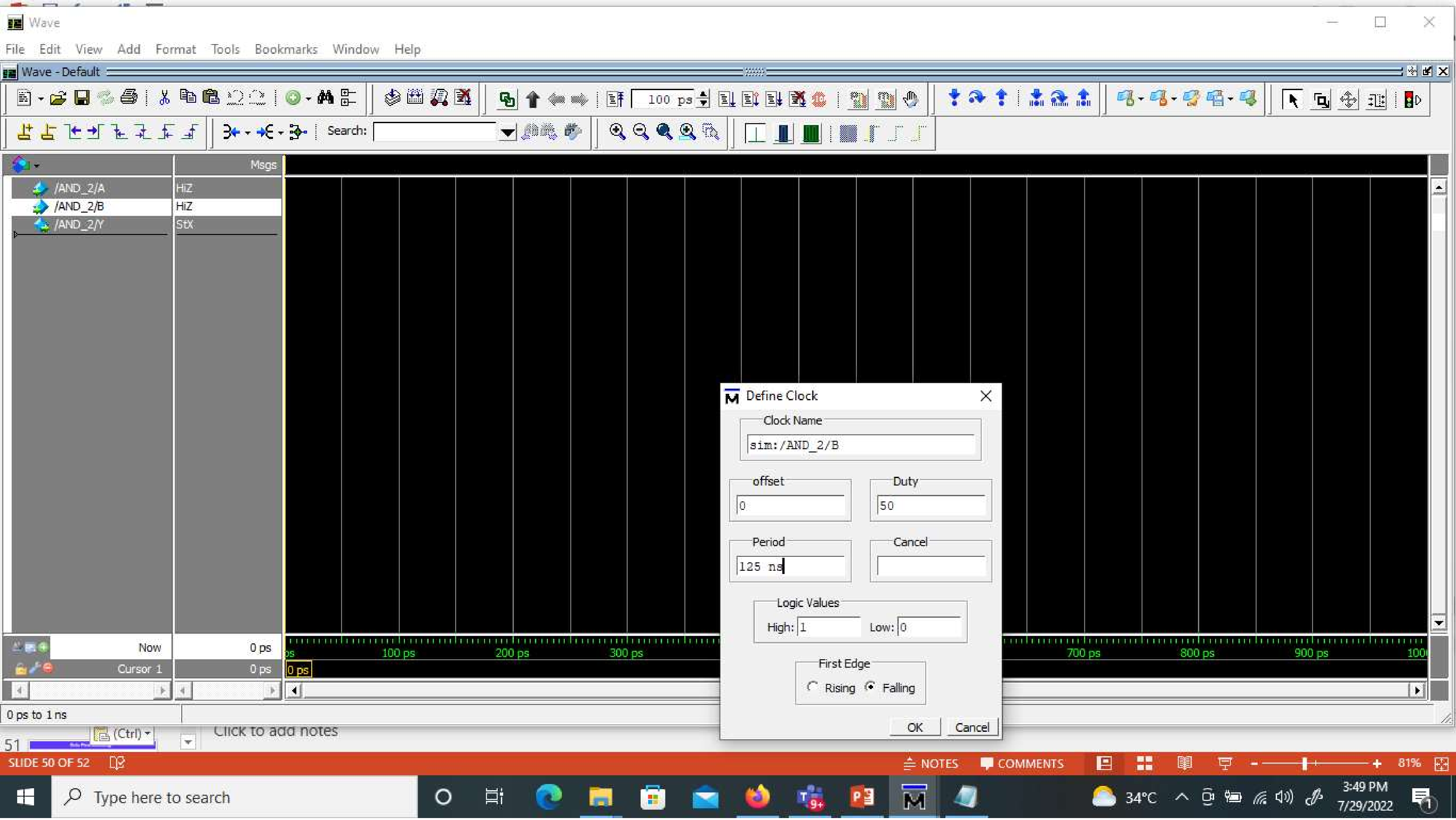

ModelSim ALTERA STARTER EDITION 10.1d
File Edit View Compile Simulate Add Objects Tools Layout Bookmarks Window Help
100 ps
Layout Simulate
ColumnLayout Default
sim - Default
Instance Design unit Design unit type Visibility Total coverage
AND_2 AND_2 Module +acc=<...
#AND#4 AND_2 Process +acc=<...
#vsim_capacity# Capacity +acc=<...
Objects
Name Value Kind Mode
A HiZ Net In
B HiZ Net In
Y StX Net Out
Project : Gate_Level_Modeling
Now: 0 ps Delta: 0
sim:/AND_2
Type here to search
34°C
3:44 PM
7/29/2022



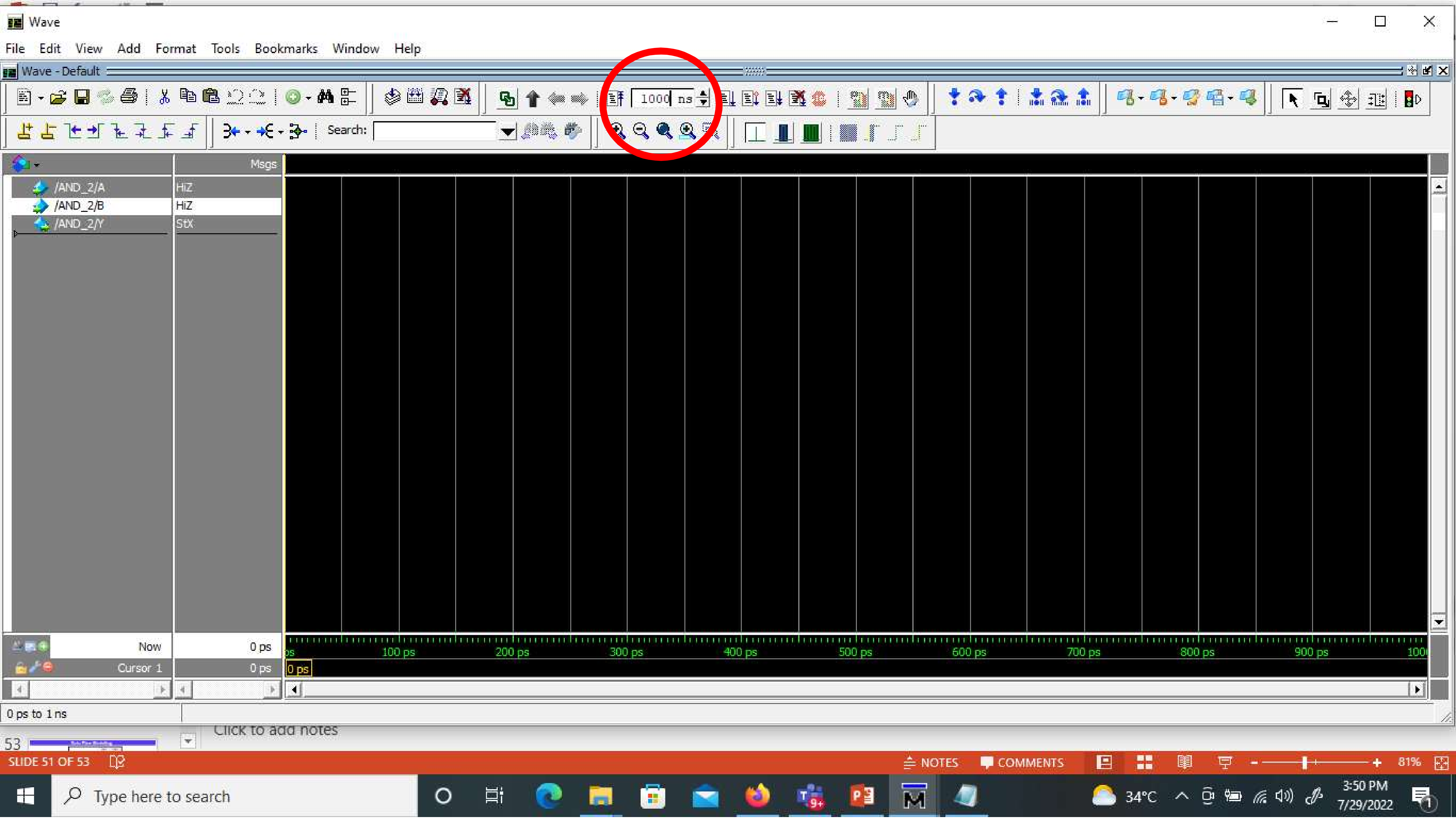
ModelSim ALTERA STARTER EDITION 10.1d
File Edit View Compile Simulate Add Objects Tools Layout Bookmarks Window Help
Layout Simulate
ColumnLayout Default
sim - Default
Instance | Design unit | Design unit type | Visibility | Total coverage
AND_2 AND_2 Module +acc=<...
#AND#4 AND_2 Process +acc=<...
#vsim_capacity# Capacity +acc=<...
Objects
Name | Value | Kind | Mode
A HiZ Net In
B HiZ Net In
Y
View Declaration
View Memory Contents
Add Wave Ctrl+W
Add Wave New
Add Wave To
Add Dataflow Ctrl+D
Add to
Copy Ctrl+C
Find... Ctrl+F
Insert Breakpoint
Toggle Coverage
Modify
Radix...
Show
Wave
List
Log
Dataflow
Watch
Selected Signals
Signals in Region
Signals in Design
Project : Gate_Level_Modeling
Now: 0 ps Delta: 0
sim:/AND_2
Type here to search
34°C
3:45 PM
7/29/2022


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/AND_2/A
/AND_2/B
/AND_2/Y
Object Declaration
Add
Edit
View
Radix
Format
Combine Signals...
Group...
Ungroup
Force...
NoForce
Clock...
Properties...
Now
0 ps
Cursor 1
0 ps
100 ps
200 ps
300 ps
400 ps
500 ps
600 ps
700 ps
800 ps
900 ps
0 ps to 1 ns
Project : Gate_Level_Modeling
Now: 0 ps Delta: 0
sim:/AND_2
Type here to search
34°C
3:47 PM
7/29/2022

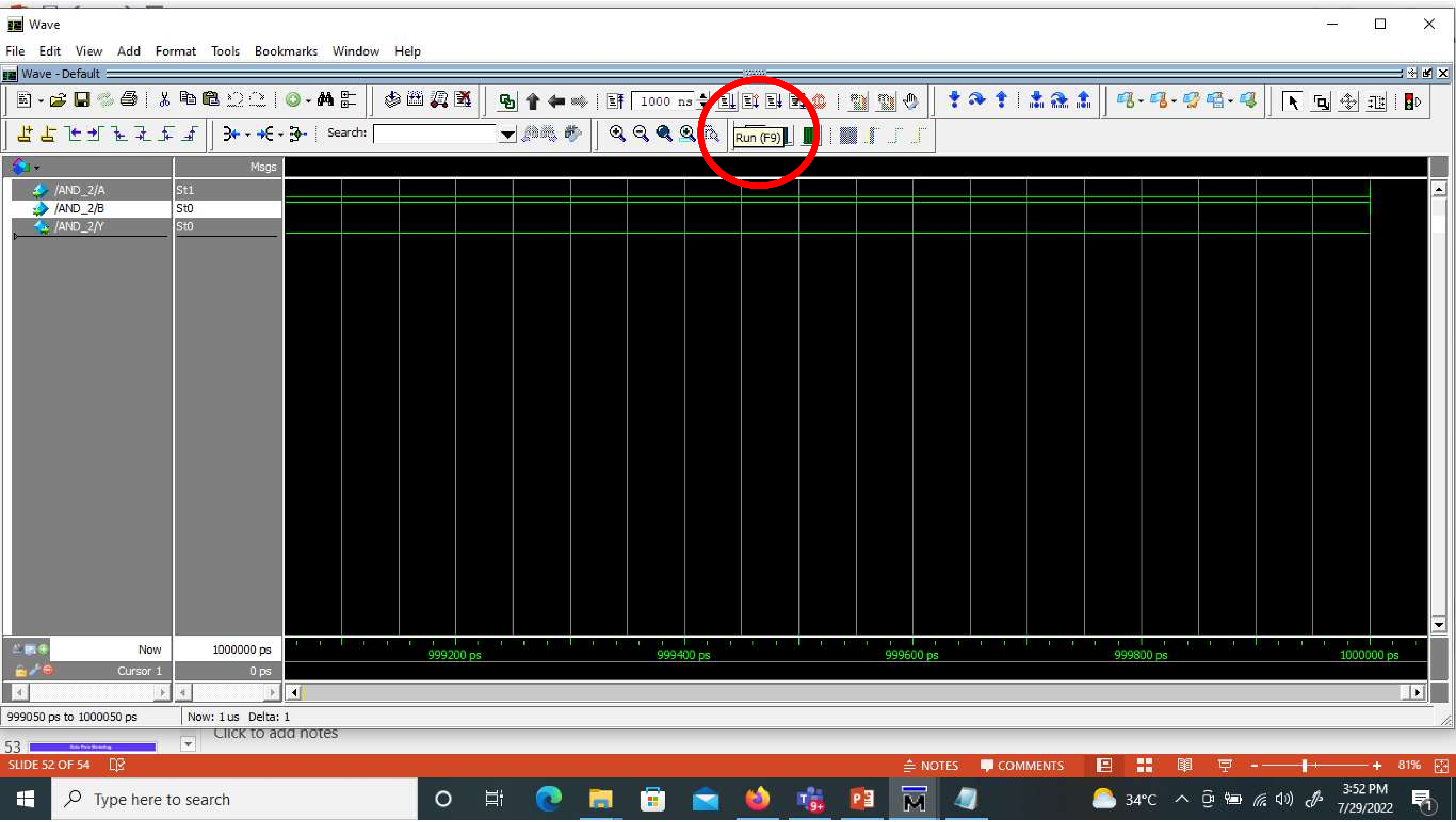

Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
100 ps
Search:
Msgs
/AND_2/A HiZ
/AND_2/B HiZ
/AND_2/Y StX
Define Clock
Clock Name
sim:/AND_2/A
offset
0
Duty
50
Period
100 ns
Cancel
Logic Values
High: 1
Low: 0
First Edge
Rising
Falling
OK
Cancel
Now 0 ps
Cursor 1 0 ps
100 ps
200 ps
300 ps
400 ps
500 ps
600 ps
700 ps
800 ps
900 ps
0 ps
0 ps to 1 ns
Click to add notes
51
SLIDE 49 OF 51
NOTES
COMMENTS
81%
Type here to search
34°C
3:48 PM
7/29/2022



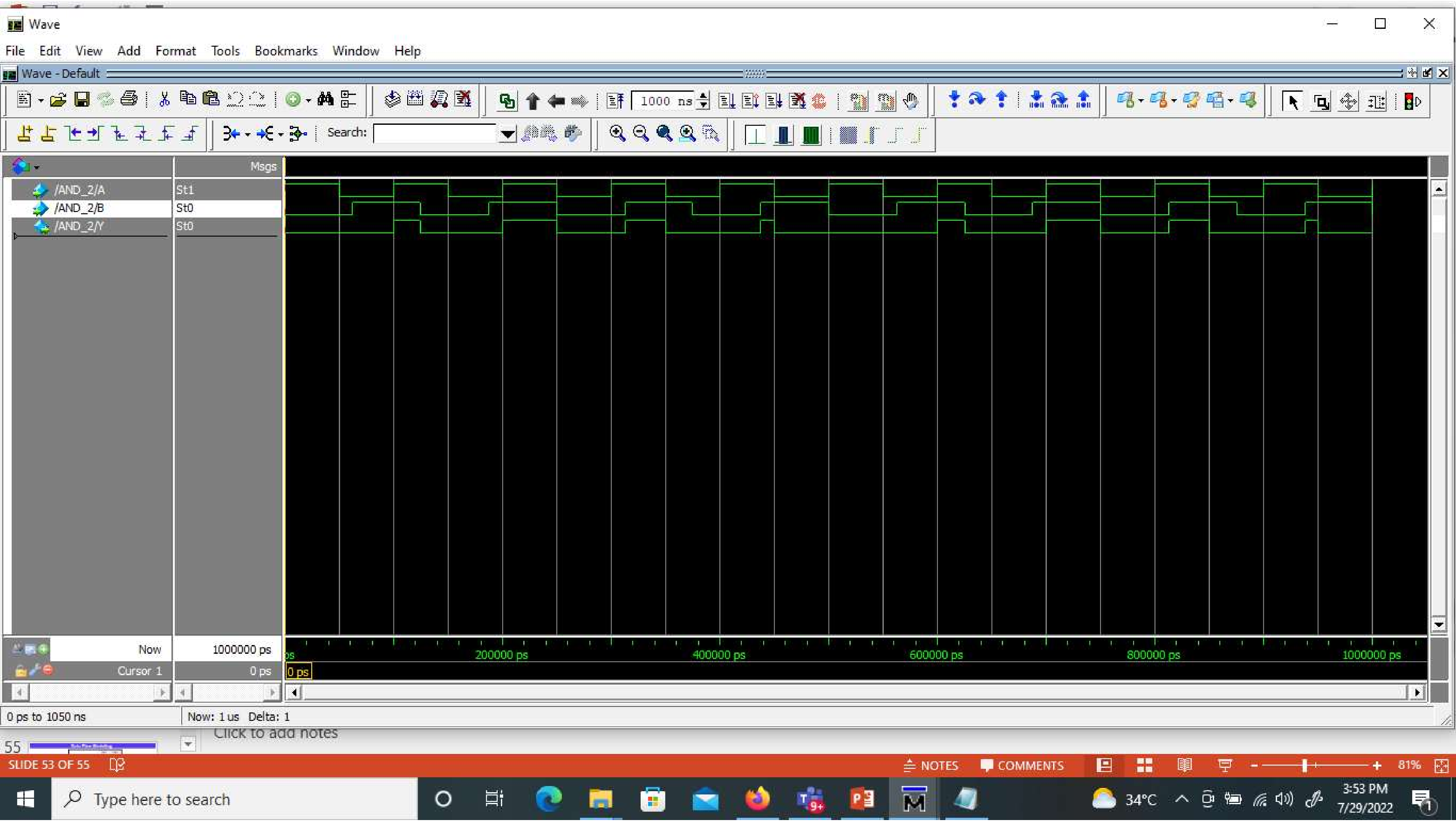
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/AND_2/A HiZ
/AND_2/B HiZ
/AND_2/Y StX
Now 0 ps
Cursor 1 0 ps
100 ps
200 ps
300 ps
700 ps
800 ps
900 ps
0 ps to 1 ns
Define Clock
Clock Name
sim:/AND_2/B
offset
0
Duty
50
Period
125 ns
Cancel
Logic Values
High: 1
Low: 0
First Edge
Rising
Falling
OK
Cancel
Click to add notes
SLIDE 50 OF 52
NOTES
COMMENTS
81%
Type here to search
34°C
3:49 PM
7/29/2022



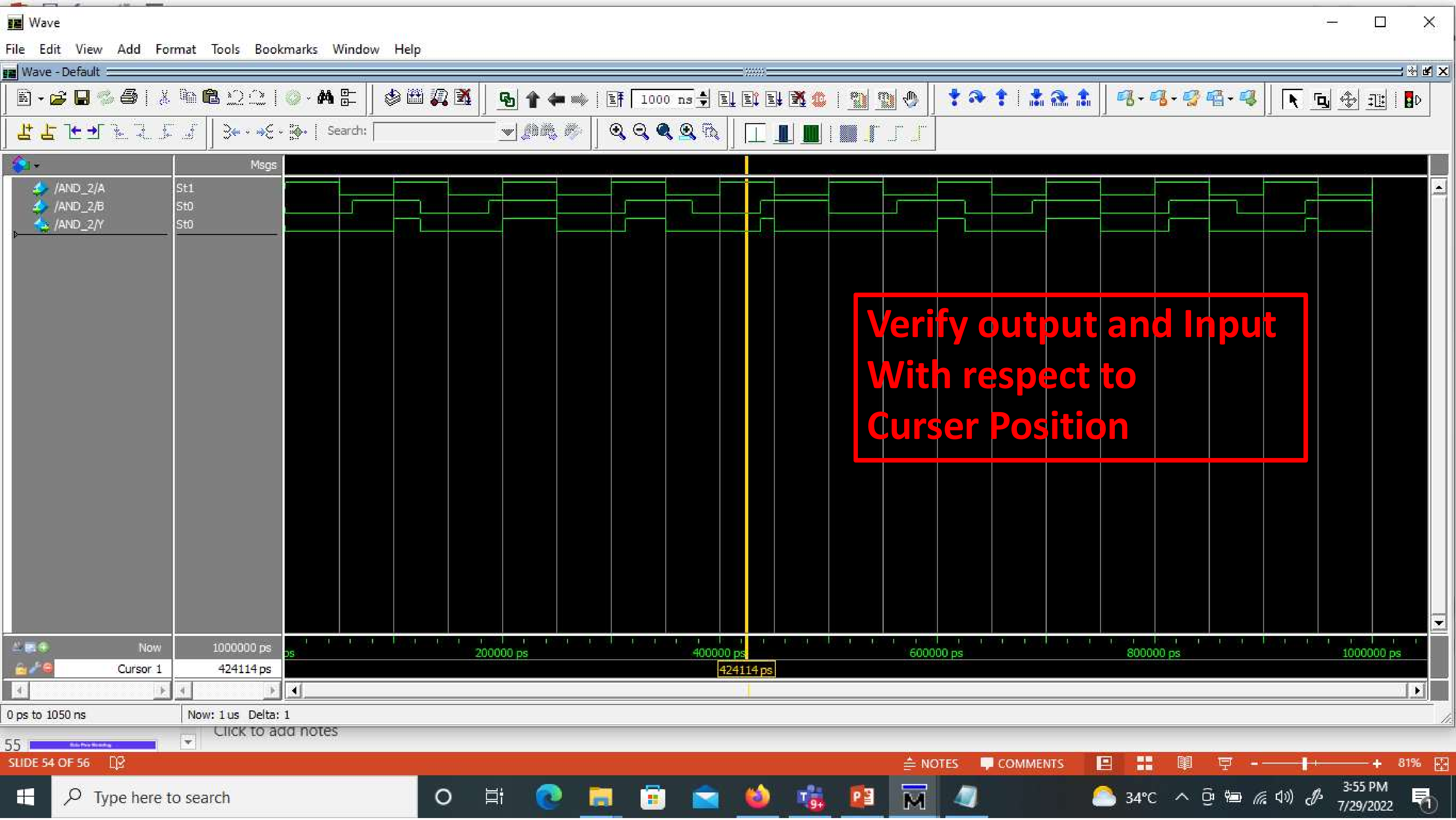
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
1000 ns
Search:
Msgs
/AND_2/A HiZ
/AND_2/B HiZ
/AND_2/Y StX
Now 0 ps
Cursor 1 0 ps
100 ps
200 ps
300 ps
400 ps
500 ps
600 ps
700 ps
800 ps
900 ps
0 ps
0 ps to 1 ns
Click to add notes
SLIDE 51 OF 53
NOTES
COMMENTS
81%
Type here to search
34°C
3:50 PM
7/29/2022



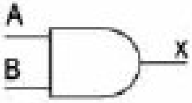
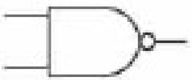
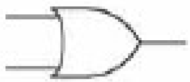
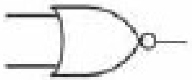
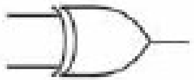
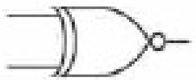
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
1000 ns
Search:
Run (F9)
Msgs
/AND_2/A St1
/AND_2/B St0
/AND_2/Y St0
Now 1000000 ps
Cursor 1 0 ps
999200 ps
999400 ps
999600 ps
999800 ps
1000000 ps
999050 ps to 1000050 ps
Now: 1 us Delta: 1
Click to add notes
53
SLIDE 52 OF 54
NOTES
COMMENTS
81%
Type here to search
34°C
3:52 PM
7/29/2022



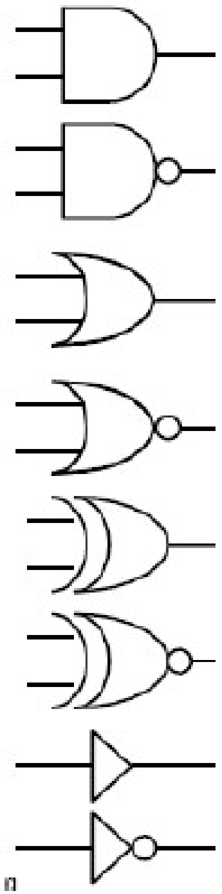
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/AND_2/A St1
/AND_2/B St0
/AND_2/Y St0
Now 1000000 ps
Cursor 1 0 ps
200000 ps
400000 ps
600000 ps
800000 ps
1000000 ps
0 ps
0 ps to 1050 ns
Now: 1 us Delta: 1
Click to add notes
SLIDE 53 OF 55
NOTES
COMMENTS
81%
Type here to search
34°C
3:53 PM
7/29/2022


Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Msgs
/AND_2/A St1
/AND_2/B St0
/AND_2/Y St0
Verify output and Input
With respect to
Curser Position
Now 1000000 ps
Cursor 1 424114 ps
200000 ps
400000 ps
600000 ps
800000 ps
1000000 ps
424114 ps
0 ps to 1050 ns
Now: 1 us Delta: 1
Click to add notes
SLIDE 54 OF 56
NOTES
COMMENTS
81%
Type here to search
34°C
3:55 PM
7/29/2022

Logic Gates

Name	NOT	AND	NAND	OR	NOR	XOR	XNOR
Alg. Expr.	$\overline{A}$	AB	$\overline{AB}$	$A+B$	$\overline{A+B}$	$A\oplus B$	$\overline{A\oplus B}$
Symbol							
Truth Table	A | X 0 | **1** 1 | **0**	B A | X 0 0 | **0** 0 1 | **0** 1 0 | **0** 1 1 | **1**	B A | X 0 0 | **1** 0 1 | **1** 1 0 | **1** 1 1 | **0**	B A | X 0 0 | **0** 0 1 | **1** 1 0 | **1** 1 1 | **1**	B A | X 0 0 | **1** 0 1 | **0** 1 0 | **0** 1 1 | **0**	B A | X 0 0 | **0** 0 1 | **1** 1 0 | **1** 1 1 | **0**	B A | X 0 0 | **1** 0 1 | **0** 1 0 | **0** 1 1 | **1**

Verilog Primitive gates



and(out, in1, in2)

nand(out, in1, in2)

or(out, in1, in2)

nor(out, in1, in2)

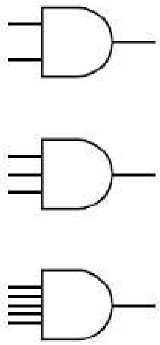
xor(out, in1, in2)

xnor(out, in1, in2)

buf(out, in)

not(out, in)

Verilog Gates (more then two inputs)



and(out, in1, in2)

and(out, in1, in2, in3)

and(out, in1, in2, in3, in4, in5)

Verilog HDL code

```
module logic_gates(Y_not,Y_and,Y_or,Y_buf,Y_nand,Y_nor,Y_xor,Y_xnor,A,B);
  input A,B;
  output Y_not,Y_and,Y_or,Y_buf,Y_nand,Y_nor,Y_xor,Y_xnor;

  // not gate
  not g0(Y_not,A);

  //and gate
  and g1(Y_and,A,B);

  //or gate
  or g2(Y_or,A,B);

  //buffer gate
  buf g3(Y_buf,A);

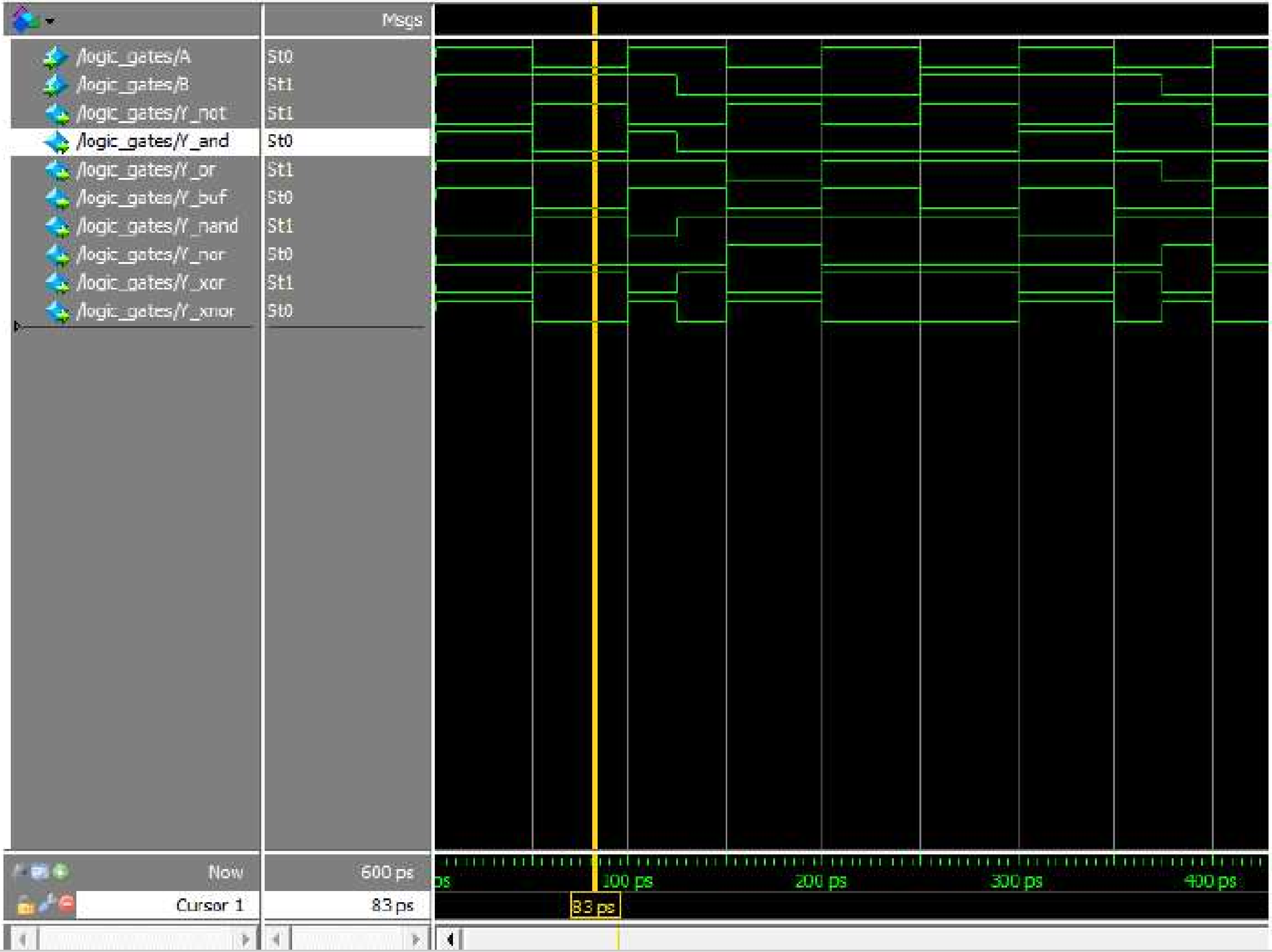
  //nand gate
  nand g4(Y_nand,A,B);

  //nor gate
  nor g5(Y_nor,A,B);

  //xor gate
  xor g6(Y_xor,A,B);

  //xnor gate
  xnor g7(Y_xnor,A,B);

endmodule
```


Msgs
/logic_gates/A St0
/logic_gates/B St1
/logic_gates/Y_not St1
/logic_gates/Y_and St0
/logic_gates/Y_or St1
/logic_gates/Y_buf St0
/logic_gates/Y_nand St1
/logic_gates/Y_nor St0
/logic_gates/Y_xor St1
/logic_gates/Y_xnor St0
Now 600 ps
Cursor 1 83 ps
ps
100 ps
200 ps
300 ps
400 ps
83 ps

Gate level Modeling

Problem statement : 1

Write Verilog code in one file for all gates using gate level modeling and then after verify its operation.

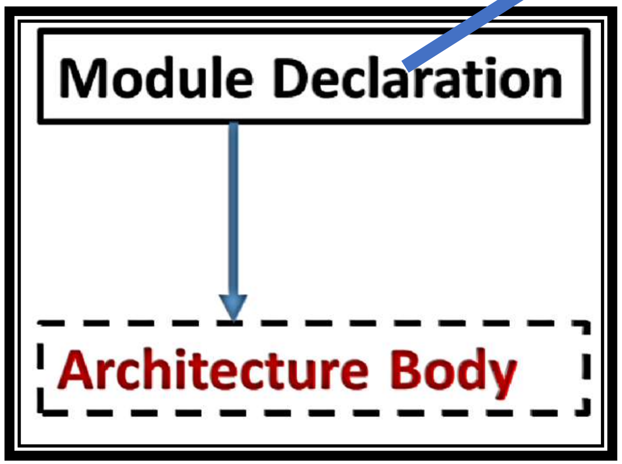
Gate level Modeling

Problem statement : 1

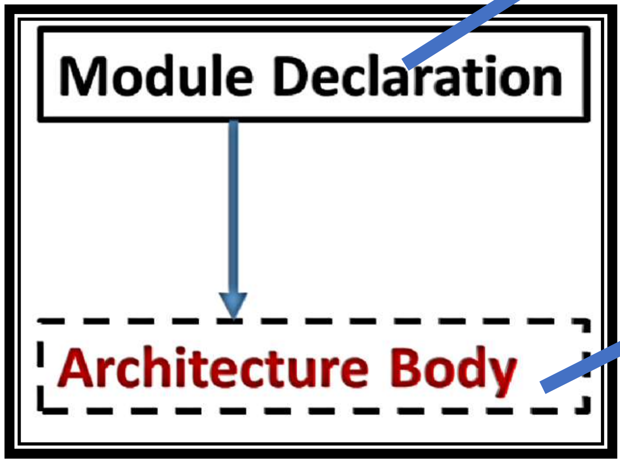
Write Verilog code in one file for all gates using gate level modeling and then after verify its operation.

Gate level Modeling

```
module  All_gate ( Y_and, Y_or, Y_not, Y_nand,
Y_nor, A, B);
 Input A,B;
 output Y_and, Y_or, Y_not, Y_nand, Y_nor;
```



Gate level Modeling

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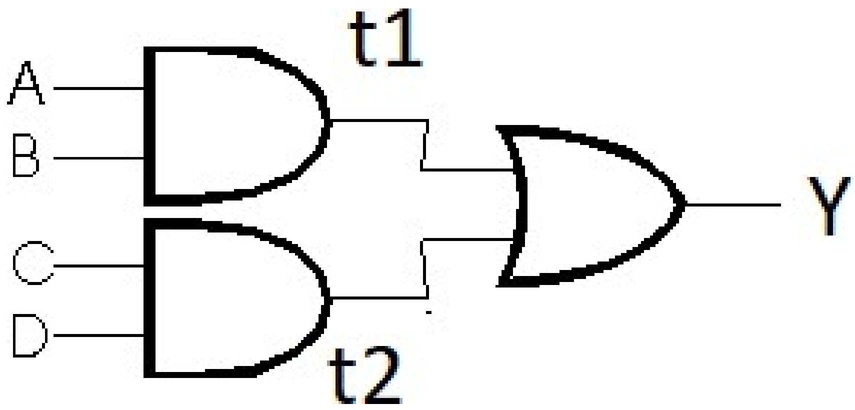
module  All_gate ( Y_and, Y_or, Y_not, Y_nand, Y_nor, A, B);
 Input A,B;
 output Y_and, Y_or, Y_not, Y_nand, Y_nor;
```

```
     and(Y_and, A, B);
     or(Y_or, A, B);
     not(Y_not, A);
     nand(Y_nand, A, B);
     nor(Y_nor, A, B);

endmodule
```

Gate level Modeling



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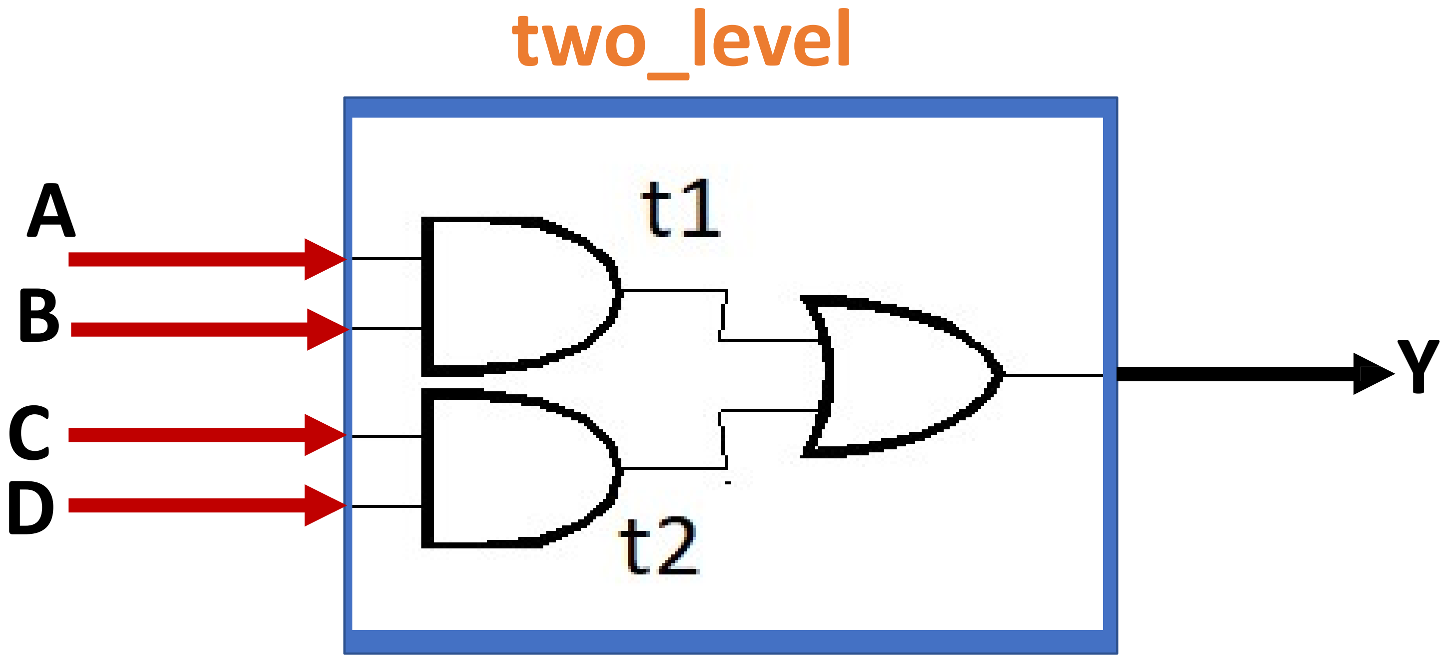

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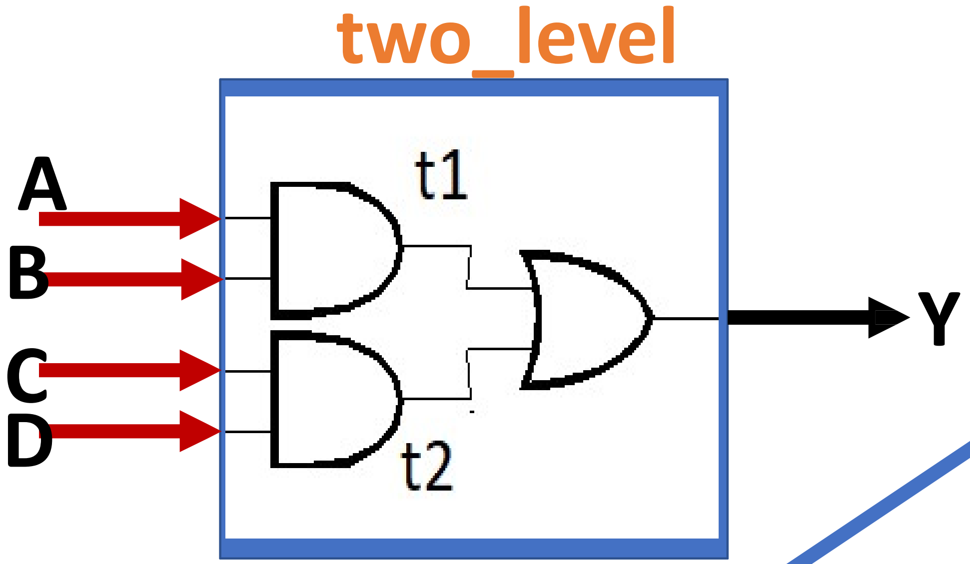
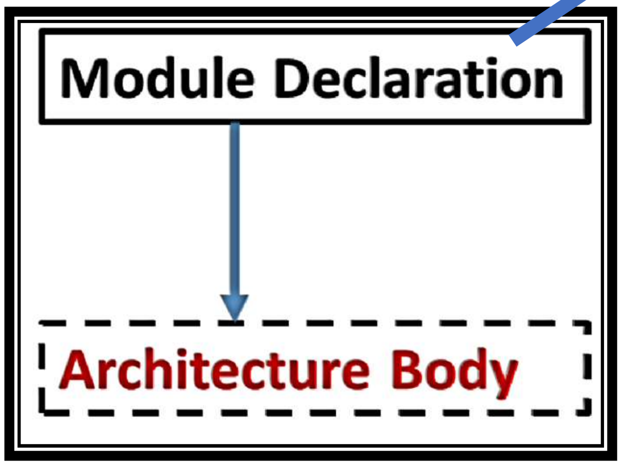



```
module  two_level ( Y, A, B,C,D);
 Input A,B,C,D;
 output Y;
```



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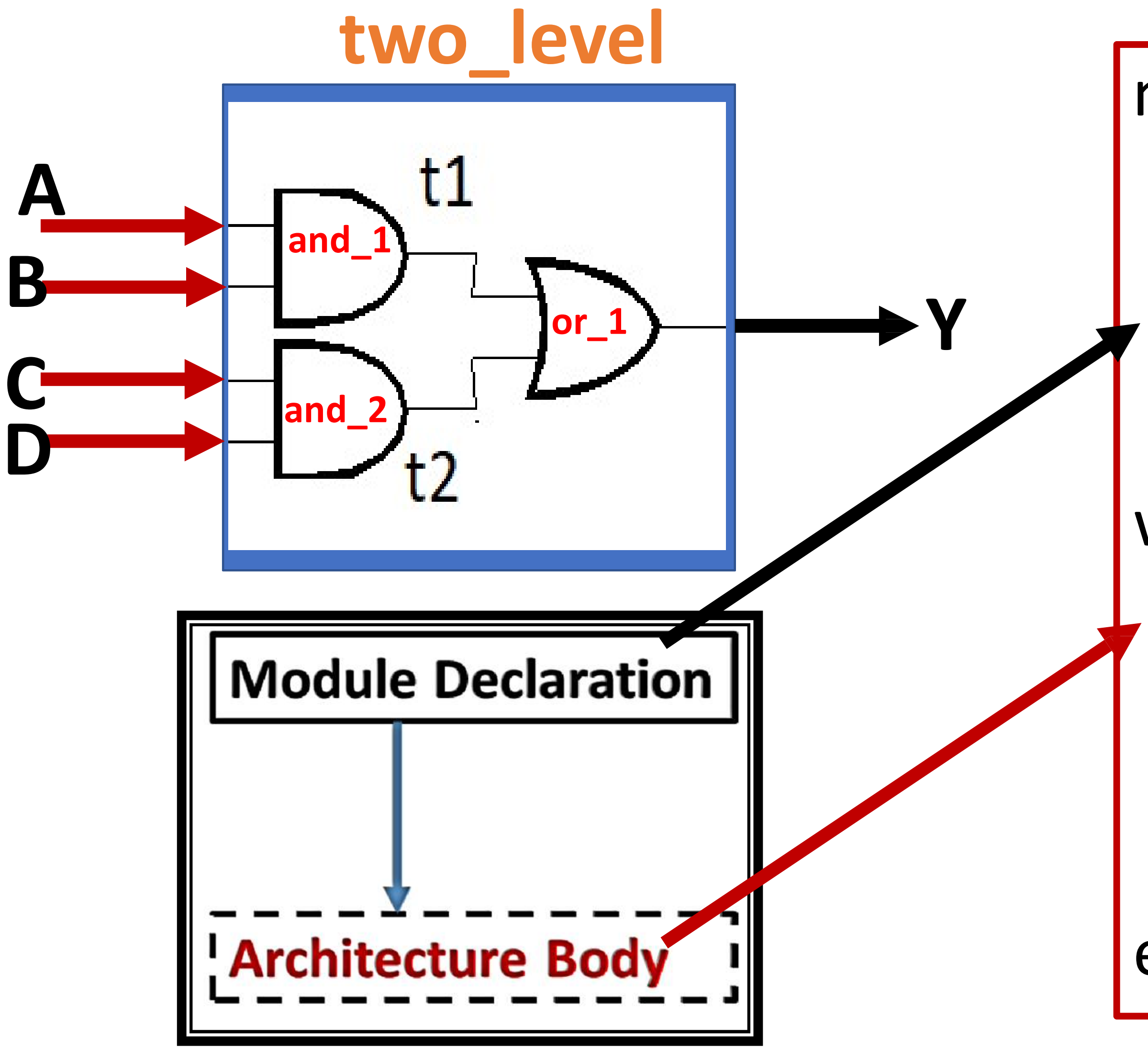



```
module  two_level ( Y, A, B,C,D);
 Input A,B,C,D;
 output Y;

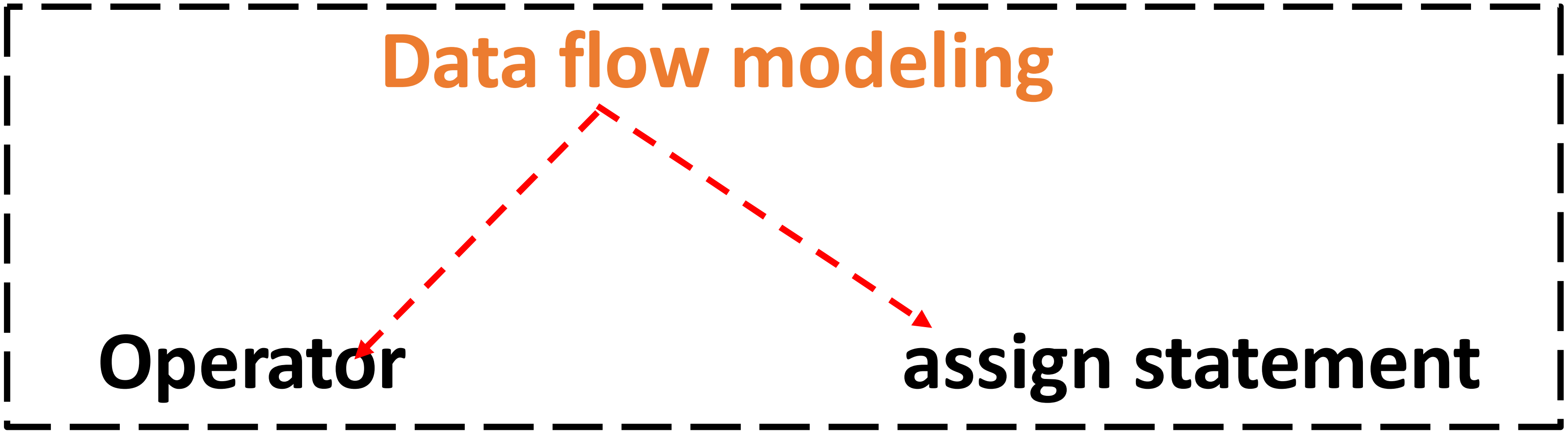

wire t1, t2;


    and and_1(t1, A, B);
    and and_2(t2, C, D);
    or or_1( t1, t2);
endmodule
```

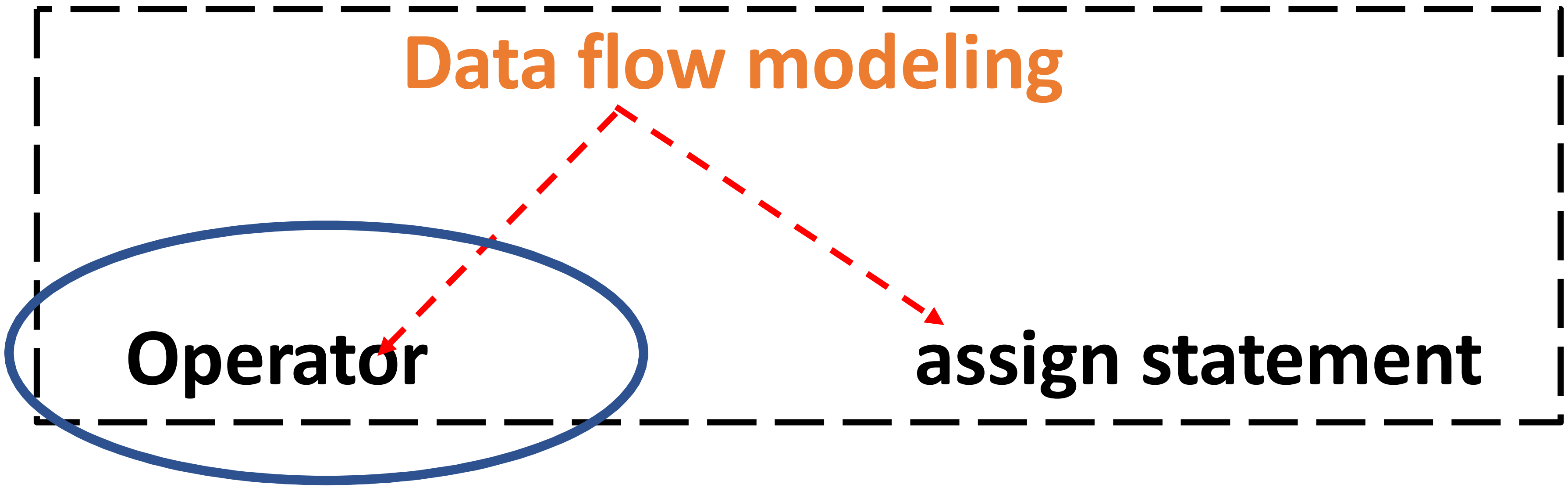
Data Flow Modeling

Data flow modeling

Data Flow Modeling



Data Flow Modeling



Verilog Operators

{ }	concatenation
+ - * / **	arithmetic
%	modulus
> >= < <=	relational
!	logical NOT
&&	logical AND
||	logical OR
==	logical equality
!=	logical inequality
===	case equality
!==	case inequality
? :	conditional

~	bit-wise NOT
&	bit-wise AND
|	bit-wise OR
^	bit-wise XOR
^~ ~^	bit-wise XNOR
&	reduction AND
|	reduction OR
~&	reduction NAND
~|	reduction NOR
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~^ ^~	reduction XNOR
<<	shift left
>>	shift right

Verilog Operators

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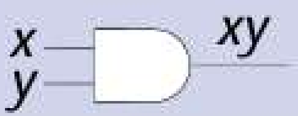
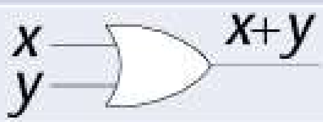
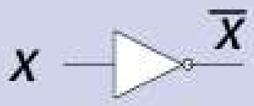
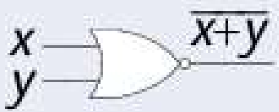
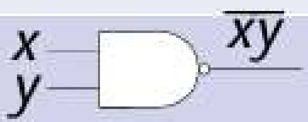
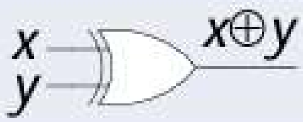
Verilog Bit wise Operator

Logical Expression	Logical Symbol	Verilog Operation
X.Y		X & Y
X+Y		X|Y
X'		~ X
(X+Y)'		~ (X|Y)
(X.Y)'		~(X.Y)
X⊕Y		(X ^ Y)
X⊙Y		~(X ^ Y)

Verilog Operators

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Verilog Operators

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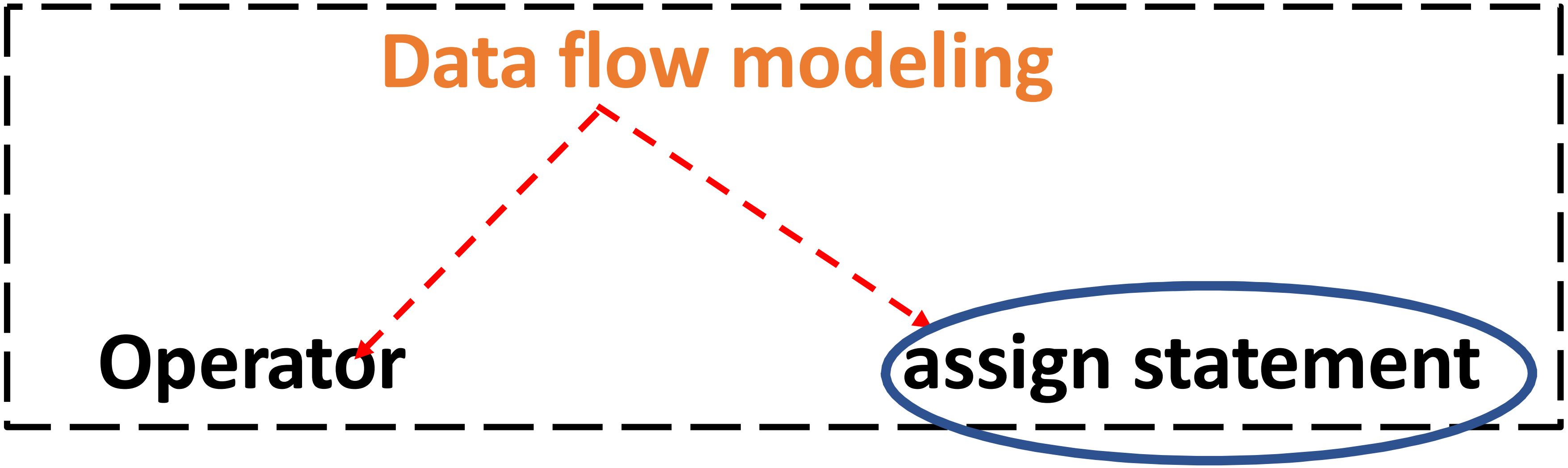
Data Flow Modeling

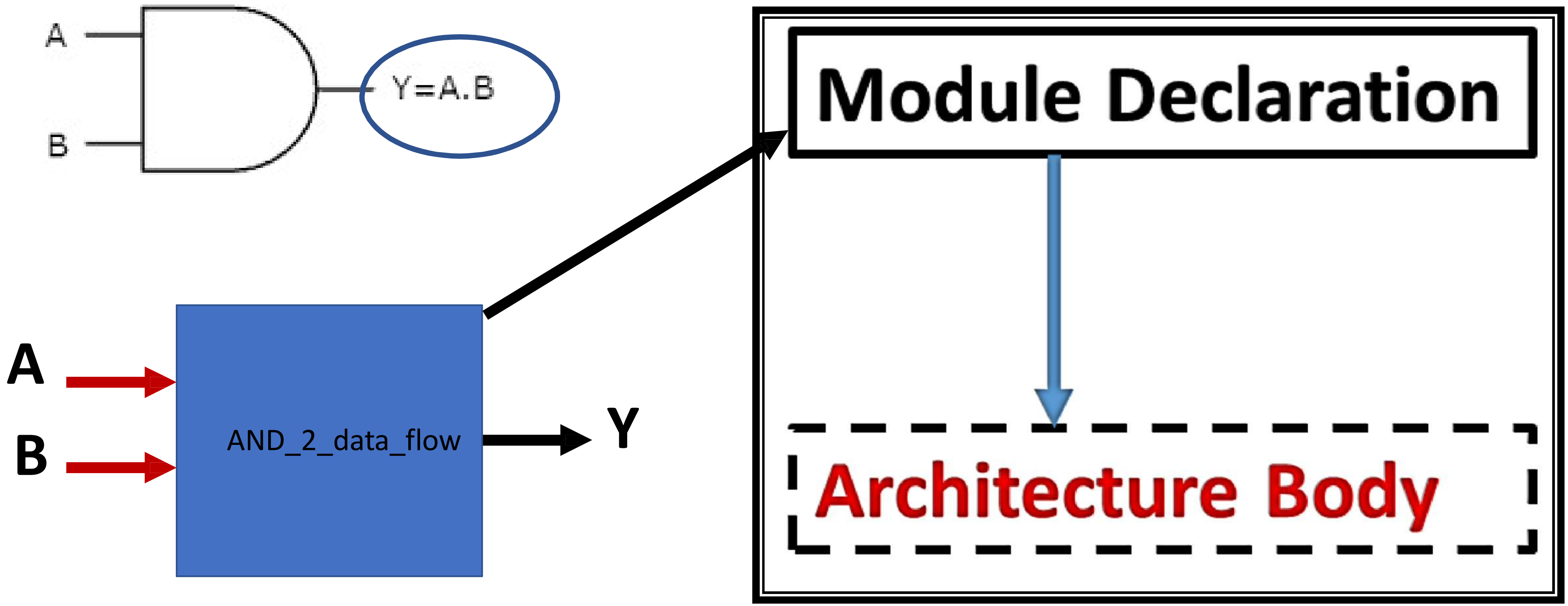


Example:
assign y = (A & B); **=> Y = A.B**

Data Flow Modeling

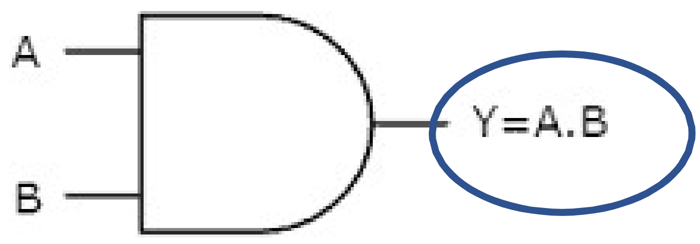
Equation of the AND gate

Data flow modeling



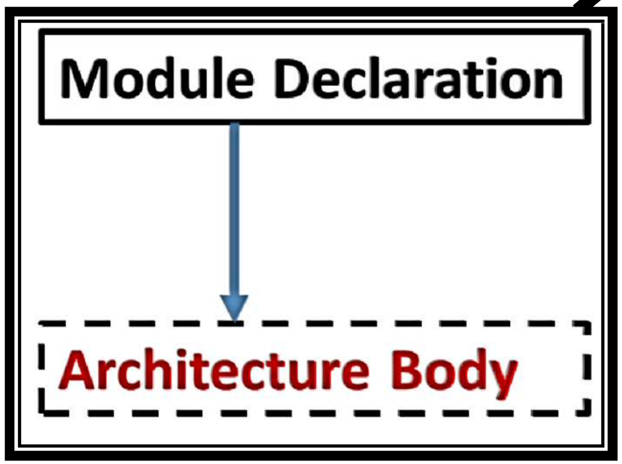
Data Flow Modeling

Equation of the AND gate



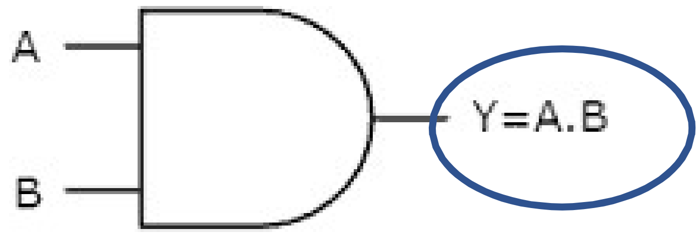
Data flow modeling

```
module AND_2_data_flow (output Y, input A, B);
assign Y = A & B;
endmodule
```



Data Flow Modeling

Equation of the AND gate



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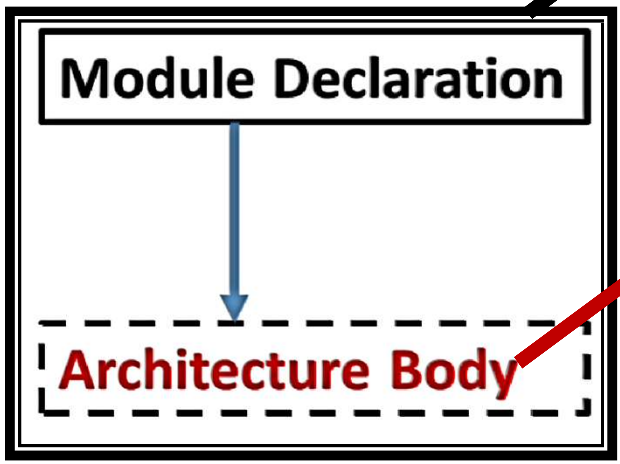


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module AND_2_data_flow (Y, A, B);
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Data Flow modeling

Problem statement : 2
Write Verilog code in one file for all gates Boolean expression using data flow modeling and then after verify its operation.

Data Flow modeling

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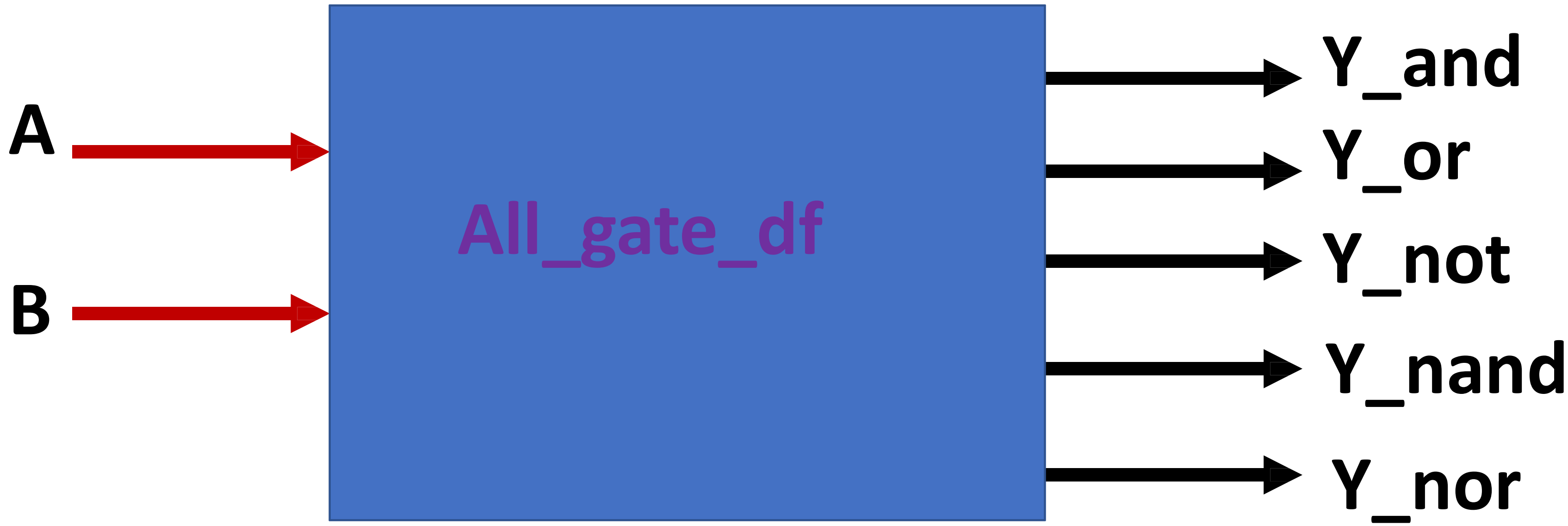
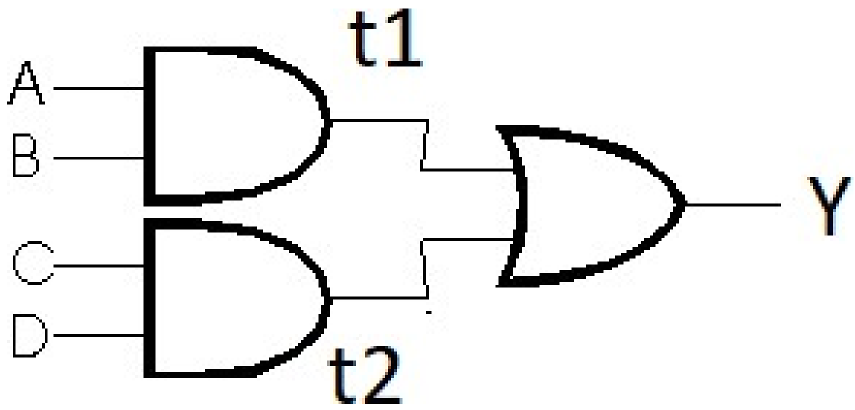
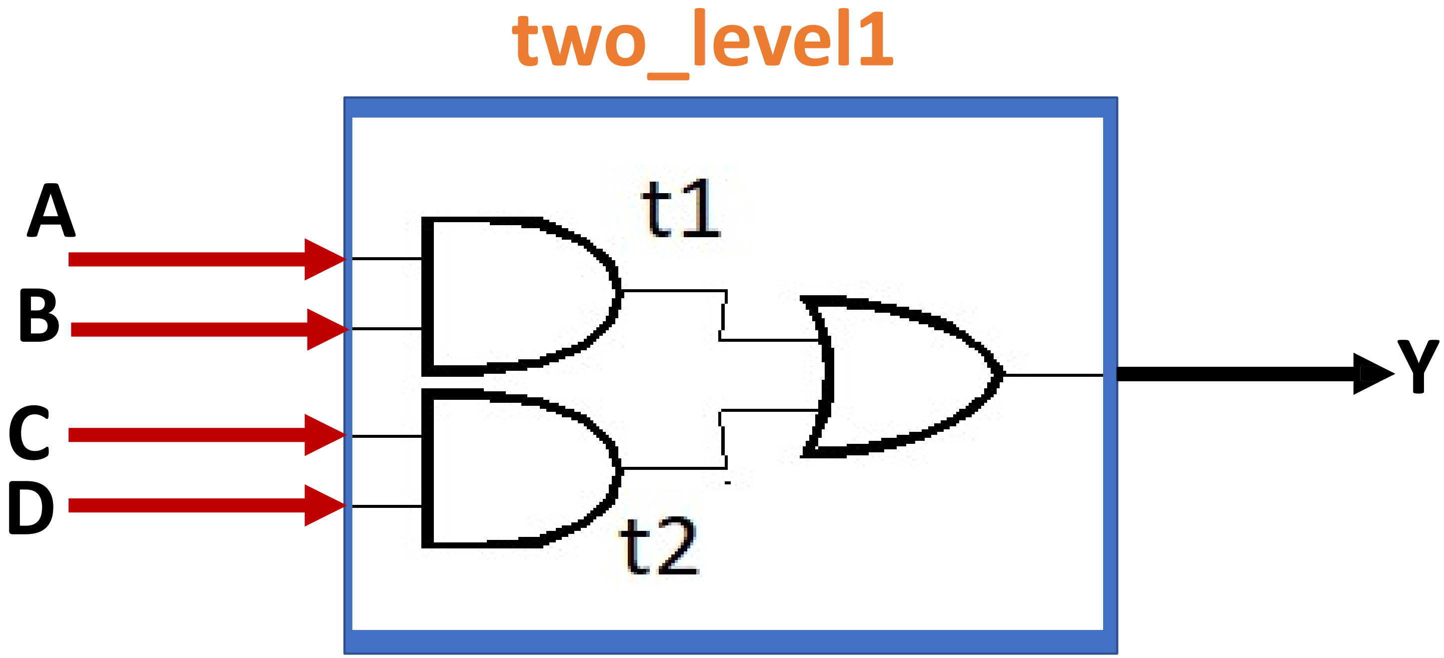


Data Flow Modeling



Assign Y= (A & B)+(C &D)

Gate level Modeling



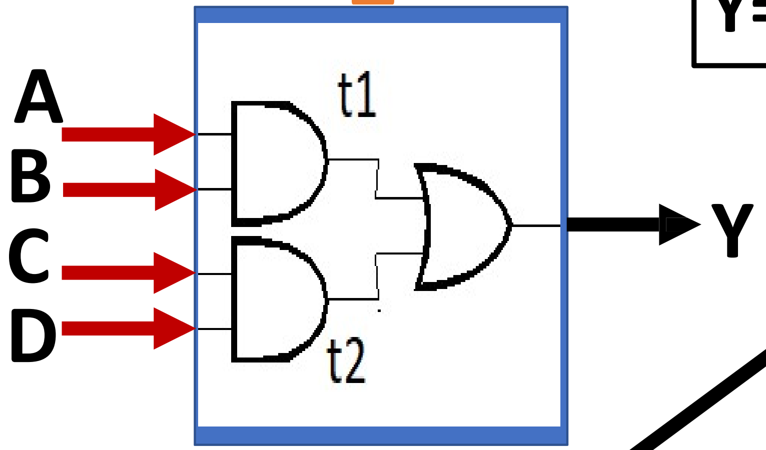
Y= AB+CD

Assign Y= (A & B)|(C &D);

Data Flow Modeling

Equation of the AND gate

two_level



Y= AB+CD

Data flow modeling

```
module  two_level1 (Y, A, B,C,D);
input A, B,C,D;
output Y;
assign Y = (A & B)|(C &D);
endmodule
```